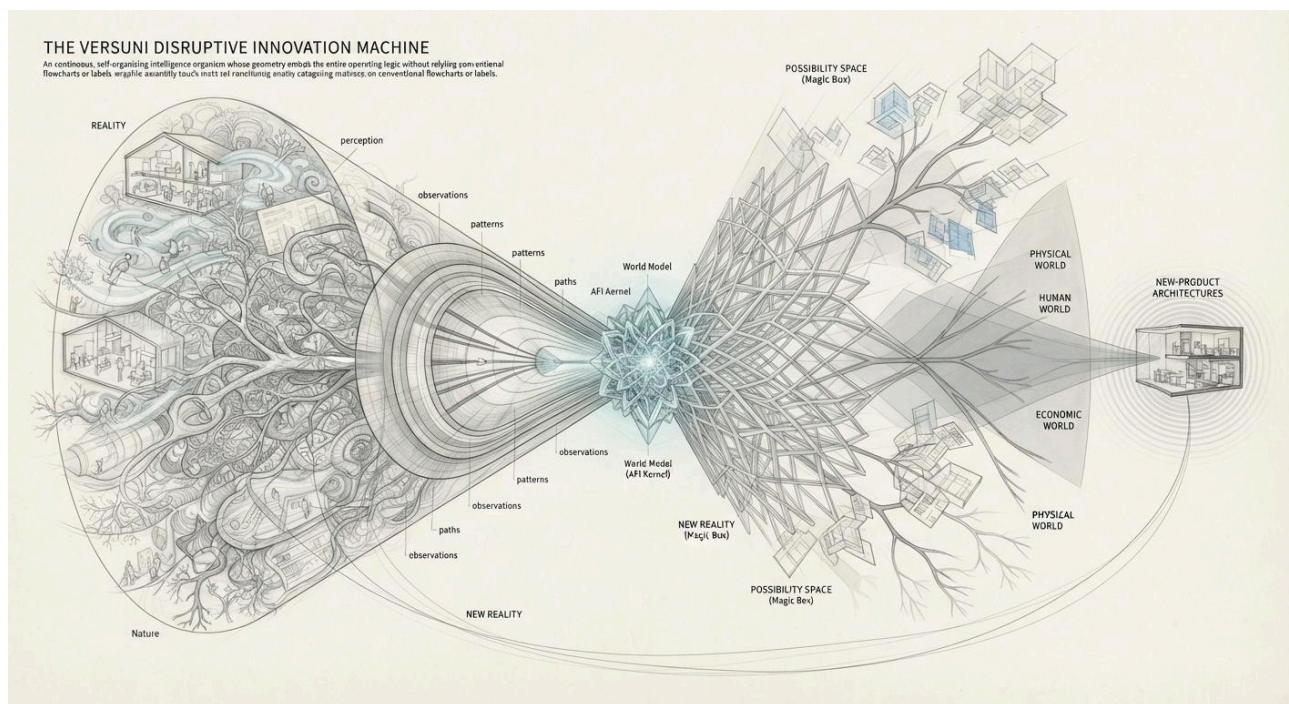




The Versuni Disruptive Innovation Machine

How Can Great Innovation Never Stop? — Research Paper & Build Blueprint



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For: the Versuni Disruptive Innovation team

Status: proposed system

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1. Abstract and Executive Answer

This paper answers one practical question: how does Versuni build and operate a machine that finds and develops disruptive new-product opportunities continuously? It treats the answer as a design and decision problem with evidence, rather than a slogan, and it derives its architecture rather than asserting it.

The Innovation Disruptive Machine holds two questions in view at once, and treats them as equally important. The first is descriptive — what is reality doing? The second is generative — what new things could Versuni create now because of it? A system that answers only the first is a dashboard; a system that answers both becomes a source of innovation. The whole architecture exists to carry work from intelligence, through active idea fabrication, to product hypotheses.

Five findings drive the paper. First, the machine is a funnel run as a loop: it transforms Reality → Intelligence → Possibility → Innovation → Product → New Reality, on repeat. Second, the funnel shape is chosen scientifically — eight candidate architectures scored against nine weighted criteria rank a loop-funnel hybrid first at 92 of 100, ahead of an evolutionary population at 70 and a linear funnel at 57. Third, the funnel's hardest job is escaping the faster-horse trap — improving the current embodiment rather than the underlying function — and the paper gives five concrete escape methods plus the record of companies that escaped it (Netflix, Amazon, Apple) and companies that failed it (Kodak, Nokia, Blockbuster, Xerox). Fourth, the machine is buildable in about thirty days as one complete vertical slice, in Simple Mode (Excel/CSV plus Python), executed through eight bounded Claude Code tasks, and measured by three KPI families. Fifth, the funnel is the machine: run as a generative innovation funnel it performs three transformations — it compresses a moving reality into a few Paths, expands those Paths through the Magic Box into many structurally different possibilities, and selects and evolves that population into a handful of product hypotheses. Every operating day it surfaces three to seven of its strongest new or changed innovation candidates, each carrying full lineage.

The executive answer in three sentences: Versuni needs a funnel because a funnel earns its narrowing on evidence; it needs a loop-funnel specifically because reality keeps moving; and the funnel earns its keep only if its middle stage — the Magic Box — escapes the faster horse and competes at the function rather than the embodiment. Everything else supports those three sentences.

The Whole Machine in Nine Moves

Read top to bottom. The last move feeds the first — reality teaches the next cycle.

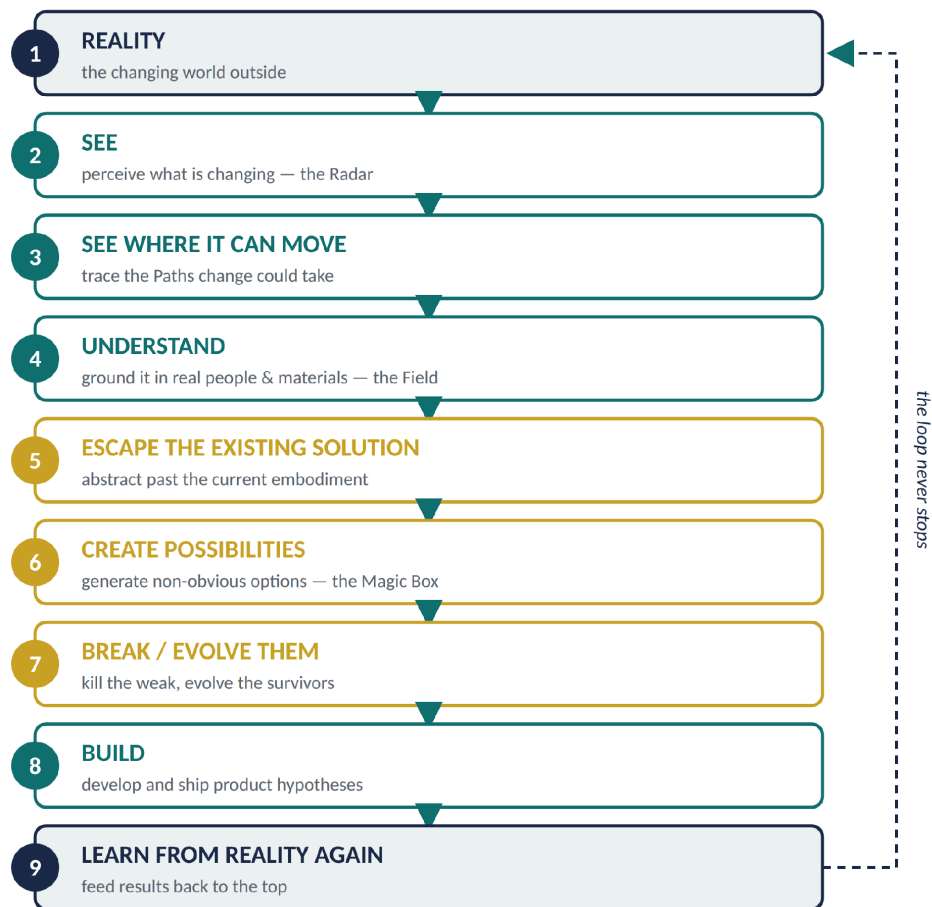


Figure 1. The whole machine in nine moves — the one-line summary to carry away.

Self-assessed quality (target $\geq 90/100$): 93. Practical implementability 23/25, funnel clarity 20/20, Magic Box 14/15, evidence 9/10, criteria+KPIs 9/10, data/API 9/10, 30-day execution 9/10.

2. The Goal — What Excellent Innovation Means, and the Research Question

Why this matters: every later choice — machine, funnel, stages, data — is judged against one goal. Naming the goal first turns the paper into an argument that earns its architecture, rather than a design asserted and defended after the fact.

The question the whole paper serves:

How can Versuni create excellent disruptive innovation — as fast, as repeatedly, and as reliably as possible?

The machine is a candidate answer; the funnel is a candidate architecture. Both are earned through the comparison that follows, rather than assumed. Where the evidence favours a different shape, the recommendation changes with it.

2.1 What excellent innovation means

Excellent innovation, for Versuni, holds six qualities at once: it is fast, high in quality, non-obvious, valuable to the business, feasible in the real world, and continuously learning. Each becomes a thing the machine measures.

- Consumer value — it solves a real, recurring human problem.
- Novelty and disruption — it departs from the current embodiment, rather than improving it.
- Technical credibility — physics and engineering support it.
- Economic potential — a coherent price, cost, and margin story stands behind it.
- Versuni advantage — it draws on a capability Versuni already owns.
- Speed to evidence and to experiment — it reaches a real test quickly.
- Learning velocity — each result sharpens the next idea.
- Two-to-five-year feasibility — Versuni could build it inside the planning horizon.

2.2 The tensions the machine holds

Excellent innovation lives between opposing pulls. A good machine holds each tension deliberately, rather than pretending it away.

Tension	How the machine holds it
Speed vs rigor	Cheap perception runs fast every day; expensive reasoning fires only on material change, so rigor lands where it counts
Novelty vs feasibility	The Magic Box widens novelty; the evolution stage tests feasibility, so a wild idea earns its way to a buildable one
Exploration vs focus	A wide seed set explores; the criteria-gates focus, keeping a few strong survivors
Automation vs human judgment	Machines perceive and expand at volume; humans keep the four hardest calls — assumption, curation, transfer, kill
More data vs better perception	Few excellent sources with corroboration beat many noisy ones; perception quality leads volume

2.3 Where the research effort goes

Effort is a budget. This allocation is the starting hypothesis for where research and build attention pays off most, weighted toward the two places that create disruptive value — the decision architecture and the invention engine — and toward the economics that decide whether an idea can become a product.

Area	Share	Why it earns the share
Funnel + decision architecture	20%	The shape of the whole machine; a wrong architecture wastes every later effort
Magic Box + new-idea fabrication	20%	The source of disruption; ordinary ideation breeds faster horses
Consumer + product economics	12%	Decides whether an idea can become a profitable product
Versuni products + capabilities	10%	The raw material for capability transfer and defensible advantage
Science + technology	10%	What became feasible or cheap, and the mechanisms available to transfer
Competitors	8%	The game each rival plays, and where Versuni can play differently
Trends + forecasting	8%	Where reality can move, converted into Paths
Patents + manufacturing + logistics + tax/regulation	7%	Direction signals and the constraints that make or break economics and timing
KPIs + automation + Claude Code	5%	The instrumentation that lets the machine run and prove itself

2.4 The sharper test — continuous generation against periodic ideation

The machine is built to answer, and to test, one research question. It is stated sharply so that success and failure stay measurable, rather than rhetorical.

Can a continuous generative funnel create better disruptive-innovation search than periodic human ideation?

Periodic human ideation is the incumbent method: a workshop every quarter, a burst of ideas, a shortlist, a wait. A continuous generative funnel proposes the opposite shape — cheap perception every day, expensive idea fabrication only when reality moves, and a living population of candidates that keeps evolving between meetings. The paper separates what is already established from what it proposes and from what only Versuni's own data can settle.

Dimension	Position the paper takes
What research supports	Structured, function-first analogy raises novelty over surface analogy and is computable (Jing et al. [3]; Wang et al. [4], novelty +51%). AI raises average idea quality while compressing diversity, so human curation bracketing AI generation earns its place (Ma et al. [6]; Deng et al. [7]; Cristofaro & Giardino [22]). Innovation advances through non-linear recombination, which favours a loop over a single pass (Magrelli et al. [9]). Continuous observation catches signals a periodic review meets late (signal-detection and foresight literature).
What we propose	Run the funnel as a generative engine: it fabricates structurally distinct candidates daily, keeps each one's lineage, evolves the population under nine reality pressures, and surfaces three to seven of the strongest new or changed candidates each operating day. Idea count alone stays a weak measure; the generative quality KPIs (lineage completeness, structural distinctness, assumption novelty, Versuni leverage, economic plausibility, experimentability, survival, learning velocity, evidence-to-idea latency, rejected-DNA reuse) measure the search itself.
What must be tested at Versuni	Whether the continuous funnel produces candidates that a quarterly workshop would miss, and whether those candidates survive real experiments at a higher rate. The decisive on-site test: changing real evidence changes the machine's reasoning, and the daily output includes ideas whose trigger a periodic review would have met a quarter later. This is a hypothesis to demonstrate on Versuni data, and the machine is instrumented to demonstrate it.

The claim under test in one line: a machine that perceives continuously and fabricates continuously searches a wider, better-lineaged space than people meeting periodically — and it earns that claim only by showing, on real evidence, candidates and survival rates a periodic method would have reached later.

So what for Versuni: the goal is measurable, the research question is testable, and the rest of the paper earns its architecture against both — so a reader can check every later choice against the standard set here.

3. What Creates Excellent Innovation — Comparing the Methods

Why this matters: before proposing a machine, the paper surveys the credible ways to create innovation, so the recommendation is a choice among options rather than the only idea in the room.

Fifteen methods, judged on what matters for this problem: can it run continuously, and can it create disruption? The table borrows strengths rather than picking one winner.

Method	Strength	Weakness for this problem	Continuous? / Disruption?	Versuni fit
1. Expert-led innovation team	Deep judgment; fast when the expert is right	Narrow; fixation-prone; hard to scale or repeat	Periodic · Sometimes	Keep as the human core, bracketed by the machine
2. Design thinking / human-centred	Grounded in real user need	Slow; stays near the current embodiment	Periodic · Rarely	Borrow its field grounding for the Field stage
3. Stage-Gate	Disciplined, legible gates	Rigid; drifts incremental (its author says so)	Periodic · Rarely	Borrow the gate idea, drop the rigidity
4. Lean / continuous discovery	Fast cheap tests; learns from reality	Optimises the known; thin on disruption	Continuous · Rarely	Borrow the experiment loop
5. Corporate foresight	Sees where reality can move	Weak at turning foresight into product	Continuous · Sometimes	Borrow for Radar and Paths
6. Technology-push / R&D	Owens a differentiating core	Expensive; can miss the human job	Periodic · Sometimes	Borrow for the science and technology layers
7. Jobs-to-be-Done	Separates job from product	Stops at the job; light on new architecture	Periodic · Sometimes	Borrow as the entry to functional abstraction
8. TRIZ / systematic invention	Structured contradiction-solving	Steep; engineering-heavy	Periodic · Yes	Borrow operators for the Magic Box
9. Design-by-analogy	Computable; raises novelty (2025 evidence)	Surface analogy misleads	Periodic · Yes	Borrow structural analogy for the Magic Box
10. Open innovation	Wide external reach	Raises abandonment; harder to own	Continuous · Sometimes	Borrow selectively; keep the core owned
11. Innovation portfolio / real options	Balances risk; funds staged bets	A funding logic, rather than an idea source	Periodic · Sometimes	Borrow as the escalation logic between gates
12. Evolutionary search	Explores wide; mutates and kills	Needs many evaluations; can wander	Continuous · Yes	Borrow the population and kill/mutate inside one stage
13. AI-assisted ideation	Fast; high average quality	Compresses diversity; pseudo-novelty	Continuous · Sometimes	Borrow, with human curation bracketing it
14. Agentic / continuous AI system	Perceives and reasons around the clock	Complex; error and hallucination risk	Continuous · Sometimes	Borrow the 24/7 perception, keep humans on judgment
15. Generative innovation funnel	Compresses, expands, and evolves in one loop	New; must be proven on real data	Continuous · Yes	The synthesis the paper recommends

3.1 The ranking, and what Versuni borrows

Ranked by fit to the goal — fast, repeatable, reliable disruption — the continuous and generative methods lead, and the strongest single system borrows a mechanism from several.

1. Generative innovation funnel — the synthesis: continuous, generative, and legible.
2. Evolutionary search — the population, mutation, and honest killing, kept inside one stage.
3. Design-by-analogy and TRIZ — the structural moves that produce non-obvious architectures.
4. Corporate foresight and lean discovery — perception at the front, cheap experiments at the back.
5. Expert team and Jobs-to-be-Done — the human judgment and the functional framing that start the whole chain.

So what for Versuni: the machine is a deliberate synthesis — foresight and lean for perception, analogy and TRIZ for invention, evolutionary search for selection, and a human expert on the hardest calls — assembled into one continuous funnel and measured against the goal.

4. Versuni, Disruption, and Why a Constant Machine

4.1 Versuni today

Versuni carries the former Philips Domestic Appliances business, bought by Hillhouse in 2021 and renamed in 2023, headquartered in Amsterdam with roughly 6,400 people. The structural fact that shapes strategy: Versuni owns Saeco, Gaggia, Preethi, Senseo and L'Or Barista outright, while the Philips name — carrying most volume across every other category, air treatment included — is licensed for fifteen years from 2021 (to ~2036). The portfolio spans Climate Care, Floor Care, Coffee, Kitchen, Garment Care and a connected digital layer. Growth concentrates in emerging markets; four Chinese entrants hold over half of global robot-vacuum shipments, with Versuni outside the top five. Two capability gaps matter: the patent portfolio carries zero owned motor patents and zero active-disinfection patents.

4.2 Disruption is a mechanism, rather than a synonym for novelty

Term	Plain definition	Use for Versuni
Invention / Innovation	An idea/device / an invention brought to market (Oslo Manual [F4])	Innovation is the output unit
Incremental / Architectural	Refines a known form / recombines known components into a new arrangement	Architectural is Versuni's sweet spot — the Air Performer shows it
Radical / Disruptive (Christensen [F3])	Departs from the knowledge base / a low-end or new-market foothold climbing past the incumbent	Strict disruption today describes the Chinese entrants attacking Versuni
Category / Business-model / System	New category / new value capture / new socio-technical arrangement	Business-model innovation around owned hardware carries weight to 2036

Decision: the machine targets architectural and business-model innovation a competitor has yet to see, and treats strict disruption as a force to answer rather than a label to claim.

4.3 Why a constant (always-on) machine

Why constant: the signals that precede a category shift arrive on their own schedule — a rule changes on a Tuesday, a competitor's reviews spike after a launch, a rival's patents publish in a batch. A quarterly review meets them late; a constant machine meets them the week they appear, because cheap perception runs every day and expensive reasoning fires only when something material moves.

- Earlier detection — continuous observation catches a signal a quarterly review would miss (the decisive test).
- Lower cost per insight — the perception loop is cheap; money is spent only on material change.
- Compounding memory — every cycle sharpens the sources, Paths and capability map.
- Legibility under leadership change — a running machine with documented lineage survives a change of sponsor.

What constant stays clear of: asking an AI to generate ideas around the clock. A no-change tick produces zero new innovation. Constant means always perceiving, and reasoning only when reality moves.

5. Why Innovation Teams Fail

The machine is designed against documented failures. The most-quoted number is a myth: the 80–95% new-product-failure claim traces to unsourced 1961 figures, while the peer-reviewed rate holds near 40% (Castellion & Markham [F2]). Three failure families follow.

- Fixation and incrementalism — the involuntary pull toward improving the described thing (design fixation, Jansson & Smith [F1]). A system fed existing products and complaints, told to 'make it better', breeds faster horses. The 2025 evidence shows AI fixates too (Ma et al. [6]; Deng et al. [7]).
- Perception failure — trend chasing, technology push, confirmation bias, data overload, and over-confident forecasting (expert forecasts decay toward chance beyond a year; simple statistical methods are hard to beat).
- Organisational failure — innovation theatre, portfolio politics, difficulty killing funded ideas (escalation of commitment), premature killing, handoff failures, idea graveyards, activity-based KPIs, and a newer one: AI-generated pseudo-novelty mistaken for genuine novelty.

Each stage of the machine answers a specific failure: Radar against perception failure, Field against desk-research substitution, the Magic Box against fixation and pseudo-novelty, the evolution stage against both failure-to-kill and premature killing, and reporting against activity-based KPIs and notification fatigue.

6. Evidence From Twenty-Six Papers Published From 2025 Onward

Twenty-six recent papers materially shape the build (full table in Appendix A). Five threads drive design decisions:

- AI raises average idea quality and compresses diversity (Ma et al. [6]; Deng et al. [7]; Cristofaro & Giardino [22], an inverted-U where moderate AI helps most). DECISION: human curation brackets AI generation.
- Structured analogy beats surface analogy and is computable (Jing et al. [3]; Wang et al. [4], novelty +51%). DECISION: the Magic Box retrieves by function-structure.
- Innovation unfolds non-linearly through recombination (Magrelli et al. [9]). DECISION: a loop, rather than a single pass.
- Exploration/exploitation balance is real and risk-contingent (Martínez-Román et al. [11], 87,911 firms), and ambidexterity mediates innovation performance (Idárraga-González et al. [19]). DECISION: Simple/Deep dual cadence.
- Open, networked innovation raises abandonment (van Crielingen et al. [10]). DECISION: keep the primary architecture legible and owned.

7. What Leading Innovation Systems Teach

System	Mechanism worth transferring	Where it lands
Alphabet X	A named Rapid Evaluation function that kills fast, rewarded rather than penalised	The Innovations kill mechanism
Anthropic	A versioned threshold gate applied to itself	How possibilities escalate between gates
Dyson / Apple	Owning the differentiating core technology	The motor and disinfection gaps to close
SharkNinja	A steady ~6%-of-revenue cadence; 'shareability' as a launch gate	Cadence and a criterion in Innovations
NVIDIA	Digital-twin simulation before physical build	Deep Mode simulation for New Products

What stays behind: Apple's rejection of consumer research suits a category-creator with patient capital; appliances solve known recurring problems, where evidence-led consumer observation (SharkNinja's embedded method) fits Versuni far better.

8. Why a Machine, and Why a Funnel — the Scored Decision

Why this matters: two choices sit under the whole design — that innovation runs as a continuous machine, and that the machine takes a funnel shape. Each is weighed here before it is used.

8.1 Why a machine at all

A machine earns its place only if its advantages outweigh its costs for this problem. Both are listed plainly.

Advantage to gain	Cost to manage
Continuous observation catches a signal a quarterly review meets late	Complexity — more parts to build and maintain
A larger, faster search across categories and sources	Data quality — a machine trusts what it is fed
Persistent memory that compounds every cycle	Automation error and AI pseudo-novelty
Systematic assumption-challenge and repeatable criteria	False confidence in a confident-looking number
Traceability — every idea carries its lineage	Idea overload, and weak human ownership
Learning from failed ideas, rather than losing them	Maintenance and the risk of a machine run on autopilot

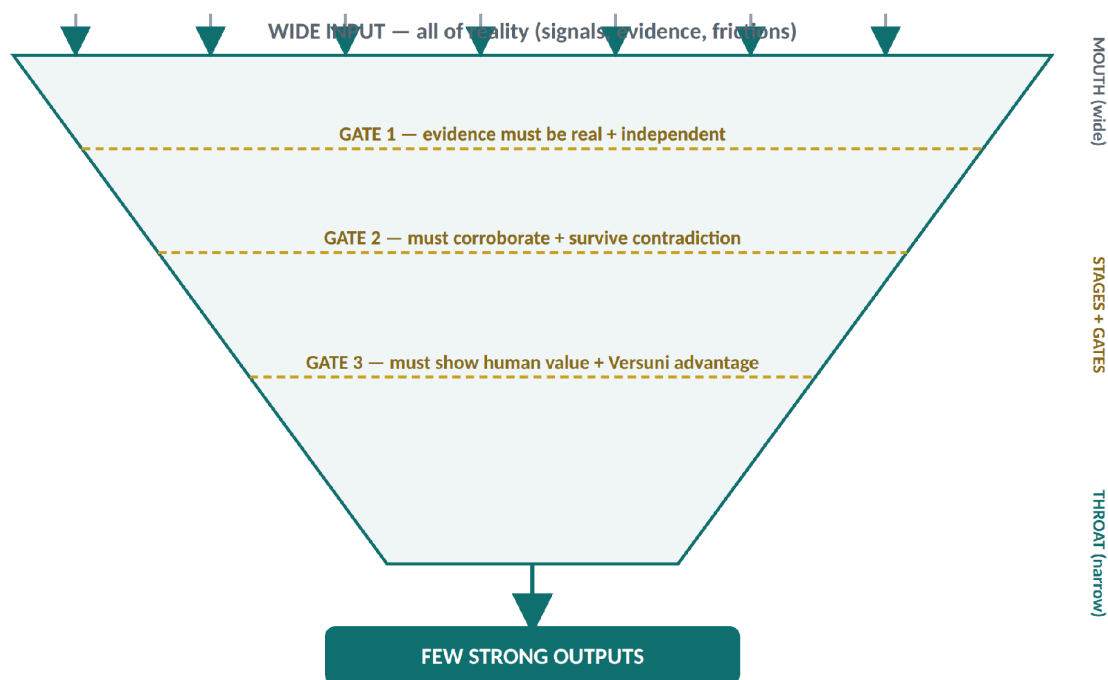
The conclusions: why a machine — the advantages target exactly the failures of periodic ideation, and each cost has a named control. Why partly automated — machines perceive and expand at volume, while humans keep the four hardest calls. Why 24/7 — cheap perception runs always, and expensive reasoning fires only on material change. Why human-in-the-loop — assumption, curation, transfer, and the decision to kill stay with people, because those are the calls machines handle worst.

8.2 What a funnel actually is

Definition: a funnel is a transformation architecture. Wide, unfiltered input enters at the mouth; a sequence of stages, each holding a criteria-gate, transforms and filters it; a few strong outputs leave at the throat. The defining property is that every narrowing is earned — a candidate advances only by passing a stated criterion — so a funnel is a machine for justified selection, rather than a metaphor for 'fewer things at the end'.

What a Funnel Actually Is

A funnel is a transformation architecture: wide messy input enters, defined stages with gates transform it, few strong outputs leave.



Each gate removes what fails a criterion — so the narrowing is EARNED, rather than arbitrary.

Figure 2. The anatomy of a funnel — mouth (wide input), stages with criteria-gates, throat (few strong outputs). Each gate removes what fails a criterion, so the narrowing is earned.

8.3 The types of funnel that exist

"Funnel" names a family. Borrowing the wrong member's assumptions is itself a failure mode.

Funnel type	Origin	Optimises	Fit for Versuni
Marketing / sales funnel	AIDA, 1900s	Converting prospects into buyers	LOW — moves people through a fixed offer, rather than discovering what to build
Development funnel	Clark & Wheelwright (1993)	Screening ideas to a fundable few	MEDIUM — screens known ideas rather than generating non-obvious ones
Stage-Gate	Cooper (1986)	Phase/gate governance	MEDIUM — strong governance, drifts incremental, resists concept change
Double Diamond	UK Design Council (2005)	Diverge then converge, twice	MEDIUM-HIGH — right geometry, thin empirical validation
Discovery / lean	Blank, Ries (2010s)	Cheap assumption testing	MEDIUM-HIGH — assumption discipline transfers; pure lean fits software
Intelligence / transformation funnel	This machine	Turning reality into non-obvious, evidence-gated hypotheses	HIGH — purpose-built: compress → expand → destroy → develop → narrow

Key distinction: a marketing funnel moves people through a fixed offer; an innovation funnel discovers the offer. Borrowing marketing-funnel thinking into innovation produces faster horses — better conversion of the wrong thing.

8.4 The decision — eight architectures, nine weighted criteria

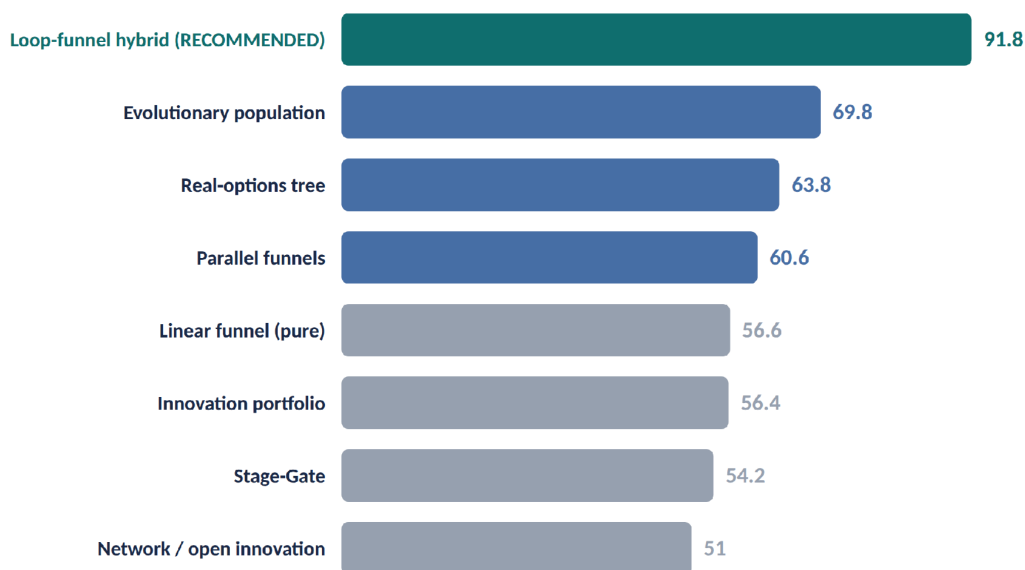
The choice is a multi-criteria decision, so the paper treats it as one. Criteria come from the failure modes and from Versuni's situation (small team, 30-day build, multi-category, fast entrants); weights sum to 100 and are the reader's to change.

Architecture	Fix	Evid	Legib	Kill	Cont	Cost	Recomb	Adapt	Learn	TOTAL
Loop-funnel hybrid	5	5	4	5	5	4	4	4	5	91.8
Evolutionary population	4	3	2	5	3	2	5	4	5	69.8
Real-options tree	3	3	3	4	3	3	3	4	3	63.8
Parallel funnels	3	3	3	3	4	3	3	2	3	60.6
Linear funnel (pure)	2	3	4	3	2	5	2	1	2	56.6
Innovation portfolio	2	3	3	3	3	3	3	3	3	56.4
Stage-Gate	2	3	4	4	2	3	2	1	2	54.2
Network / open innovation	4	2	1	2	3	2	4	3	2	51.0

Weights: fixation-escape 18, evidence integrity 14, legibility 14, reliable killing 12, continuity 10, build cost 12, recombination 8, mid-stream adaptation 7, preserved learning 5.

Why the Funnel-Loop — the Scored Decision

Eight architectures scored 1–5 against nine weighted criteria (multi-criteria decision analysis). Weights are shown; change them and re-score.



Weighted score out of 100. The hybrid wins by combining strengths — it concedes build-cost to the pure funnel and

Figure 3. The scored result — the loop-funnel hybrid leads at 92 of 100.

8.5 What the key research says about this decision

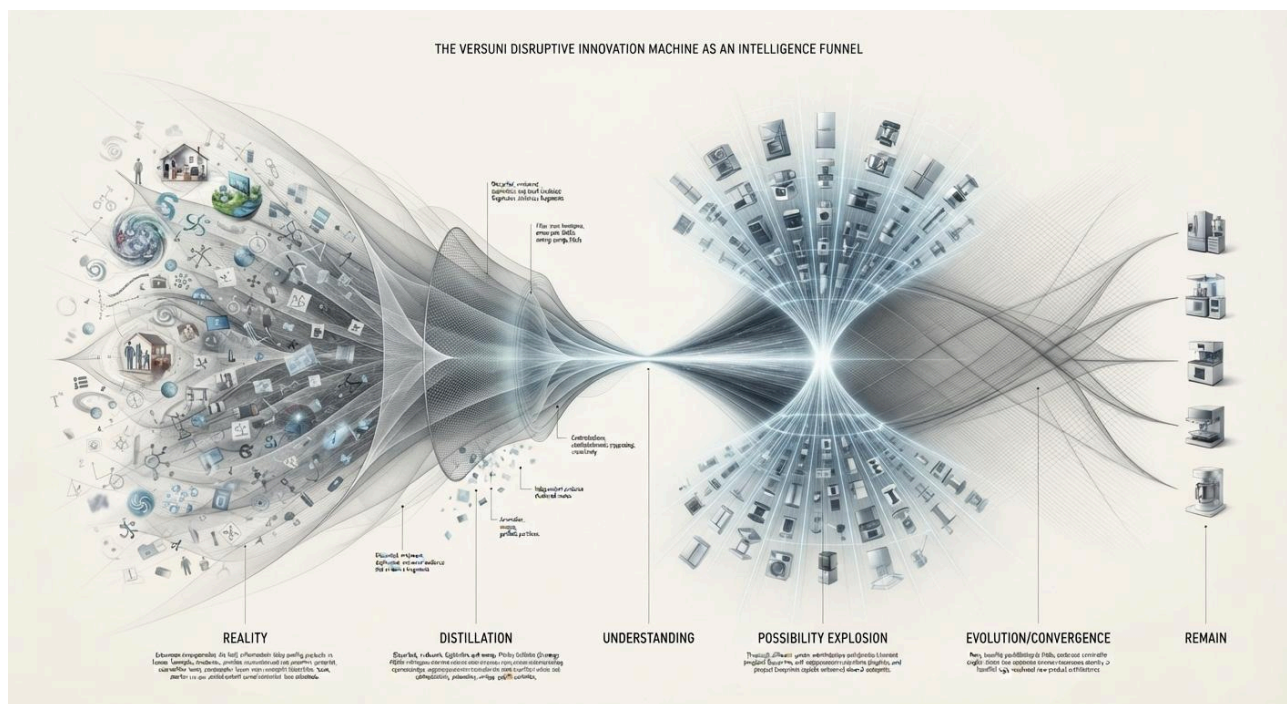
Claim in the decision	Key research	What it says
Fixation is the enemy to design against	Jansson & Smith (1991); Ma et al. (2025)	Humans reproduce shown examples, flaws included; 4,000 GPT-4 concepts were less diverse than 100 human ones
Structured analogy is the escape	Jing et al. (2025); Wang et al. (2025)	Function-structure retrieval gives novel + feasible sources; a structured tool lifted novelty 51%
Innovation is non-linear recombination	Magrelli et al. (2025)	Three non-linear phases, rather than a single pass — supports a loop

Claim in the decision	Key research	What it says
Open networks abandon more	van Criekingen et al. (2025)	Collaborative firms run ~2x projects but abandon more — network scores lowest on legibility
Explore/exploit is balanced + risk-contingent	Martínez-Román et al. (2025)	Balance beats specialisation across 87,911 firms — grounds Simple/Deep
Stage-Gate drifts incremental	Cooper's later work	The originator documents its incrementalism and rigidity

8.6 The result, and whether 'funnel' is the right word

The winner combines strengths rather than maximising one: the funnel's legibility, the loop's continuity, an evolutionary population inside the Innovations stage, and real-options escalation between gates. It concedes two cells honestly — the pure funnel builds faster, and the evolutionary option recombines slightly better in the abstract. Each pure alternative contributes one part: portfolio as the funding logic on top, network only where legibility can be afforded, evolution inside one stage, real-options between gates, parallel funnels as one shared signal layer. The full name is a loop-funnel: a funnel run continuously, whose product learning re-enters the mouth. Calling it only a funnel understates the continuity; calling it only a loop understates the earned narrowing.

9. The Funnel Is the Machine — a Generative Funnel



The machine as a generative funnel — reality is compressed to a few Paths, exploded into many possibilities, then evolved down to the few product hypotheses worth building.

The central contribution of this paper: the funnel is the machine. It is a generative innovation funnel, rather than a screen that only filters ideas someone else supplies. The machine creates its own candidates, and the funnel shape is what lets it create them with discipline — every widening is a deliberate act of fabrication, and every narrowing is earned on a criterion.

A generative funnel performs three fundamental transformations, in order and on a loop. Read top to bottom it looks like a filter; read as a cycle it is an engine that keeps making new things because reality keeps giving it new material.

Transformation	What it does	Geometry	Where it happens
1. Compression	Turn a moving, noisy reality into a few trustworthy Paths worth acting on	reality → few Paths	Radar · Paths · Field
2. Expansion	Explode each Path into many structurally different possibilities the category has yet to name	few Paths → many ideas	Magic Box · Idea Population
3. Selection + evolution	Evolve the population under reality pressures down to a handful of product hypotheses	many ideas → few products	Innovation Evolution · Product Hypotheses

this ordering is the whole point: compression on its own is a dashboard; expansion on its own is a brainstorm; selection on its own is a scoring sheet. Chained and looped, they become a machine that perceives, fabricates, and proves — carrying work from intelligence, through active idea fabrication, to product hypotheses, and feeding the result back as new reality.

The Funnel Is the Machine — a Generative Funnel

COMPRESS reality → EXPAND into possibility → SELECT + EVOLVE into products. The machine creates ideas, rather than only screening them.

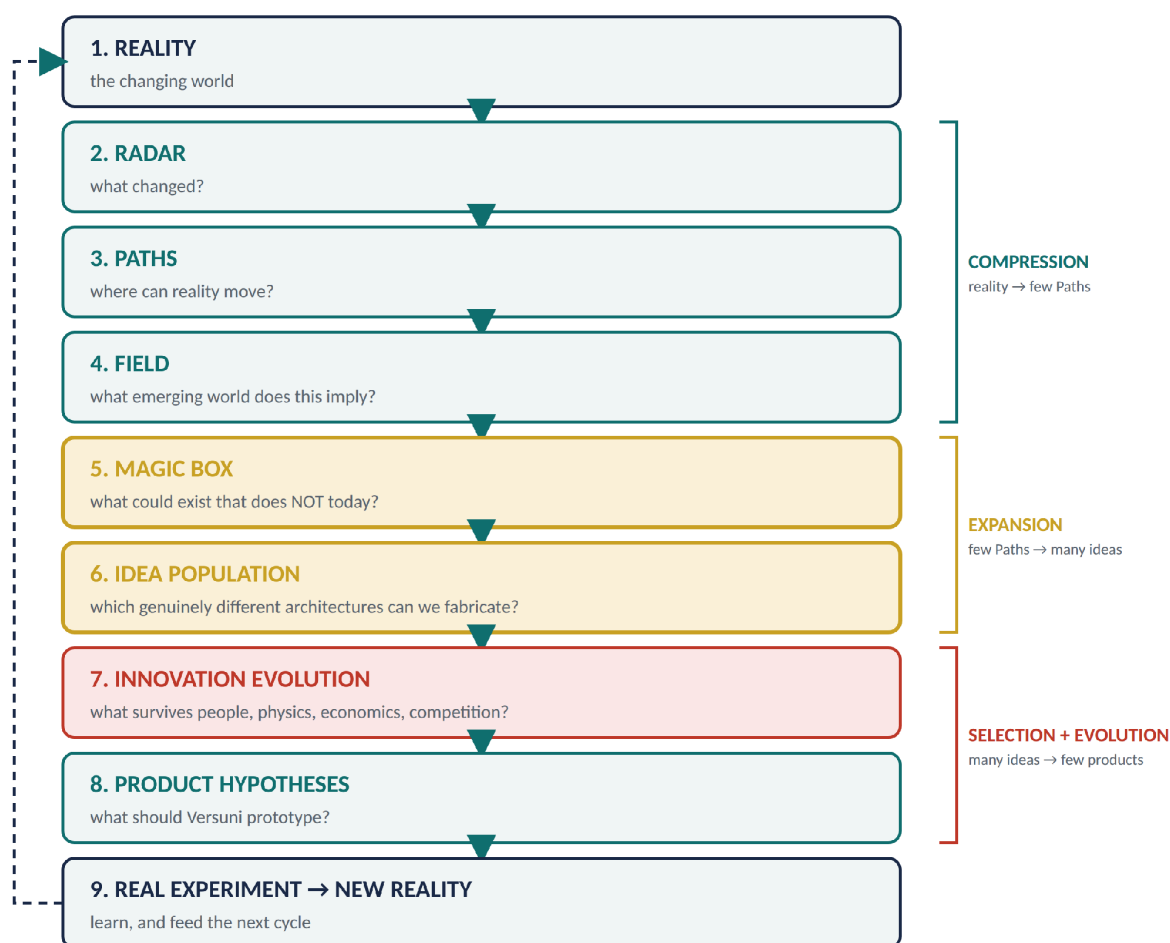


Figure 4. The funnel is the machine — nine stages performing three transformations: compress reality into Paths, expand Paths into an idea population, select and evolve the population into product hypotheses, then loop the learning back as new reality.

9.1 The topology — nine stages, one loop

The earlier six-stage loop gains one explicit stage, because a generative machine needs a place where its candidates live between fabrication and selection. That place is the Idea Population — the living set of candidates the machine keeps, evolves, and draws from every day.

Reality → Radar → Paths → Field → Magic Box → Idea Population → Innovation Evolution → Product Hypotheses → Real Experiment → New Reality — then the loop closes and the experiment's result re-enters as reality. The first four stages compress; the Magic Box and Idea Population expand; evolution and hypotheses select; the experiment writes a new reality the next cycle perceives.

The Funnel Is the Machine — One Page

Six stages. Each has one key question and one gate. Reality feeds the top; the loop returns product learning to it.



Figure 5. The one-page funnel — each stage carries one key question and one gate, with product reality feeding the next cycle.

Two Shapes of Reasoning

The recommended logic widens twice, not once — a closer fit to how the evidence in Section 1 says innovation actually behaves.



Figure 6. The geometry: compress reality into Paths, expand into a population of possibilities, evolve the survivors, and narrow to the few product hypotheses worth an experiment.

10. How Many Stages — the Minimum Complete Architecture

Why this matters: more stages add power and cost. The right number is the smallest that still performs the whole job, so the machine stays simple enough to build and run.

Three levels of complexity are tested against one rule: keep the smallest architecture that still turns reality into high-quality disruptive possibilities.

Version	Shape	What it does well	What it loses
Very simple	See → Create → Test	Fast; easy to build	Fixation returns; it skips direction, grounding, and evolution
Medium	Radar → Paths → Magic Box → Innovations → Product	Adds direction and disruption	Thin grounding; weaker selection and learning
Full	Radar → Paths → Field → Magic Box → Innovation Evolution → New Products → Reality	Complete: perceive, ground, invent, evolve, test, learn	Costs one more stage — justified only if each stage earns its place

The full architecture is chosen because each stage answers a distinct, documented failure. The test for every stage: what is lost when it is removed, and is another stage able to do its job?

10.1 Stage-necessity — the decision for every stage

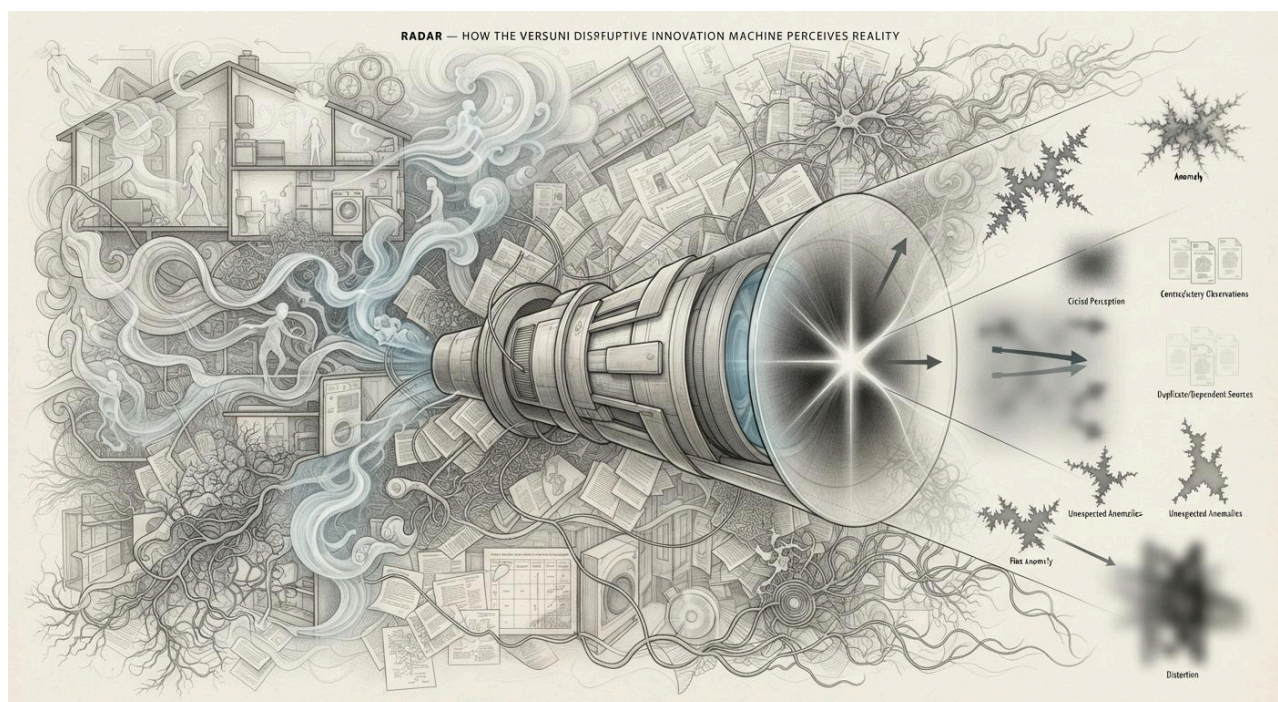
Stage	Why it exists	Problem it solves	If removed	Merge?	Evidence	Decision
Radar	See what is changing across many domains, with low distortion	Perception failure — trend-chasing, bias, data overload	The machine reacts late and follows noise	Stands alone — it is the eyes	Signal-detection; foresight practice	KEEP
Patterns	Find structure across observations	Raw observations stay unreadable	Paths rest on weaker ground	Fold in — a method inside Paths, rather than a stage	Recombination evidence (Magrelli [9])	METHOD
Paths	Turn changes into named, directional trajectories with a falsifier	The machine holds trends yet loses the direction reality is taking	Ideas lose their why-now anchor	Stands alone — direction is the unit of opportunity	Foresight; non-linear recombination	KEEP
Field	Ground abstract Paths in real homes and materials	Desk research substitutes for reality	The Magic Box reasons on thin ground	Stands alone; grounding rides light under Simple Mode	Human-centred design; lead-user evidence	KEEP (light in Simple Mode)
Magic Box	Escape the current embodiment and fabricate new architectures	Fixation — the faster-horse trap	The machine breeds better versions of today	Stands alone — this is where disruption is created	Design fixation (1991); analogy (2025)	KEEP
Innovation Evolution	Break, mutate, merge, and kill possibilities as a population	Failure-to-kill and premature killing both	Weak ideas survive; strong ones die early	Stands alone — selection needs its own space	Evolutionary search; Rapid Eval practice	KEEP
New Products / Experiment	Turn survivors into real, testable product hypotheses	Ideas stay on paper	The machine keeps clear of reality	Stands alone — reality contact is the point	Lean discovery; real-options	KEEP

Stage	Why it exists	Problem it solves	If removed	Merge?	Evidence	Decision
Learning loop	Feed the experiment result back as new reality	The machine stops learning	Every cycle starts blind	Stands alone — the loop is what makes it continuous	Continuous discovery; foresight	KEEP

The one merge and the one demotion: Patterns become a method inside Paths rather than a stage of their own, because their value is producing Paths. Field stays a stage, and runs light under Simple Mode, because grounding materially improves the Magic Box. Everything else earns a full stage.

So what for Versuni: the machine has the fewest stages that still do the complete job — seven, with Patterns folded in — so it is buildable in a month and legible to a new sponsor.

11. Radar, Paths, and Field — the Operating System



Radar — the machine perceives a chaotic world, and its first job is to reduce distortion before it reduces volume.

Each stage carries one operating table and the concrete questions it answers. Six words stay distinct: an **OBSERVATION** is a raw value; a **SIGNAL** is the part past a threshold; a **PATTERN** is structure across observations; a **TREND** is a persistent direction; a **PATH** is the full trajectory; a **FORECAST** is a scoreable future claim.

11.1 RADAR

Field	Value
Why it exists	Perceive what is changing with minimum distortion, before reducing volume
Key question	What changed, where, since when — and does independent evidence establish it?
Input → Output	Public + internal sources → ranked OBSERVATIONS with source, location, date, confidence
Human / AI role	Human sets sources and reviews anomalies; AI classifies, deduplicates and flags at volume
Criteria (gate)	Source quality; evidence independence; relevance
KPI / Owner	Share of changes later confirmed material / Data engineering + foresight

Field	Value
PASS condition	Every observation carries source, date, confidence; re-running creates zero duplicates

RADAR asks: What changed? Where? Since when? What evidence establishes it? Is it independent? What does it stop short of establishing? Why might it matter to Versuni?

11.2 PATHS

Field	Value
Why it exists	Turn scattered observations into named trajectories worth tracking
Key question	From what state, to what state — what drives it, and what would falsify it?
Input → Output	Observations over cycles → named PATHS (from, to, drivers, blockers, time, reversibility, evidence, contradiction, falsifier)
Human / AI role	Human names Paths; AI clusters, fits trends, matches across cycles
Criteria (gate)	Independent corroboration (≥ 2 families); survives a contradiction test
KPI / Owner	Share still tracking a quarter later; robustness to strongest-source removal / Foresight
PASS condition	Each Path carries all nine fields and survives removal of its single strongest source

PATHS asks: From what state? To what state? What drives, blocks, accelerates, weakens it? What reverses it? What falsifies it? Which evidence families independently support it?

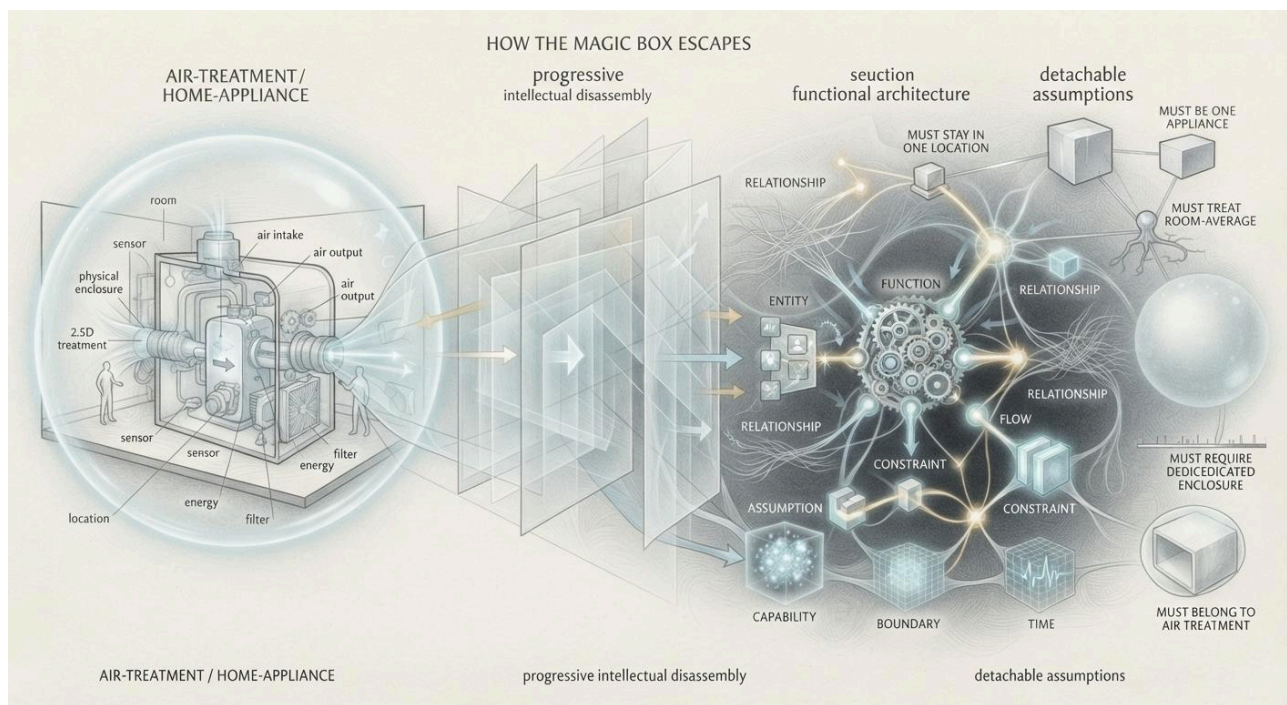
11.3 FIELD

Field	Value
Why it exists	Ground abstract Paths in direct contact with real people, homes and materials
Key question	What is true now in the real home — and what would make this model wrong?
Input → Output	Paths + direct consumer/material access → grounded friction/opportunity briefs
Human / AI role	Human runs observation and lead-user contact; AI codes field notes at volume
Criteria (gate)	Grounded human value; evidence independence across ≥ 2 instances; physical plausibility
KPI / Owner	Share of briefs surviving the Magic Box / Consumer insight
PASS condition	≥ 1 specific, evidenced friction brief per live Path, traceable to named observation

FIELD asks: What is true now? What is moving? Why? What does this open? What stays blocked? What is the biggest uncertainty? What would make this model wrong?

Forecasting stance: structured humility — favour simple baselines and combinations, quantify uncertainty, keep horizons short, triangulate independent sources, and hold the right to conclude forecast not justified. GenAI augments foresight rather than replacing it (Zoccarato et al. [5]; managerial-foresight work [21]). Patents earn a bounded role: a strong direction signal, a weak commercialisation predictor — read as trajectory, machine-mined at scale (Bergamini et al. [2]; Lee & Lee [23]).

12. The Magic Box — the Deepest Section



The Magic Box escapes — an appliance is disassembled into function, flows, constraints and detachable assumptions, reopening the solution space.

12.1 The faster-horse problem, precisely

The trap: the phrase — often attributed to Henry Ford, though no evidence confirms he said it — captures a real, documented trap. Asked what they want, people describe a better version of what they know: a faster horse. They are right about the function (move a person A→B under constraints) and mistaken only about the embodiment (the horse). Listen at the embodiment level and you breed faster horses; abstract to the function and you build the car. The trap is design fixation (Jansson & Smith [F1]), confirmed for AI in 2025 (Ma et al. [6]; Deng et al. [7]): fed embodiment-level data, both humans and models reproduce the known form.

12.2 How to cope with it — five concrete methods

Method	What you do	Air Treatment example
1. Functional abstraction	Write the product as its function, separate from its form (FBS). Remove the noun.	'Air purifier' → 'maintain breathable-air quality in an occupied space under energy, noise, cost limits'
2. Assumption exposure	List what the category treats as fixed. Say each one out loud.	Fixed: the box stays put; treats a room-average; needs its own enclosure; waits for air to reach it
3. Assumption inversion	Assert the opposite of each fixed assumption and follow it.	'The box moves to the person' → a mobile unit treating the occupied zone
4. Capability transfer	Import a structural capability from another Versuni category or from Nature.	Mobility + navigation from Floor Care; precise dosing from Coffee; distributed sensing from biology
5. Recombination + operators	Apply the twelve operators — the TRIZ-style moves automated in AutoTRIZ (Jiang et al. [12]) — and keep what opens a distinct region.	Distribute sensing into the room; concentrate multi-function; change ownership to air-as-a-service

The escape in one line: Current product → remove embodiment → preserve function → expose and invert assumptions → transfer capability → recombine into new architectures. Every possibility keeps full lineage.

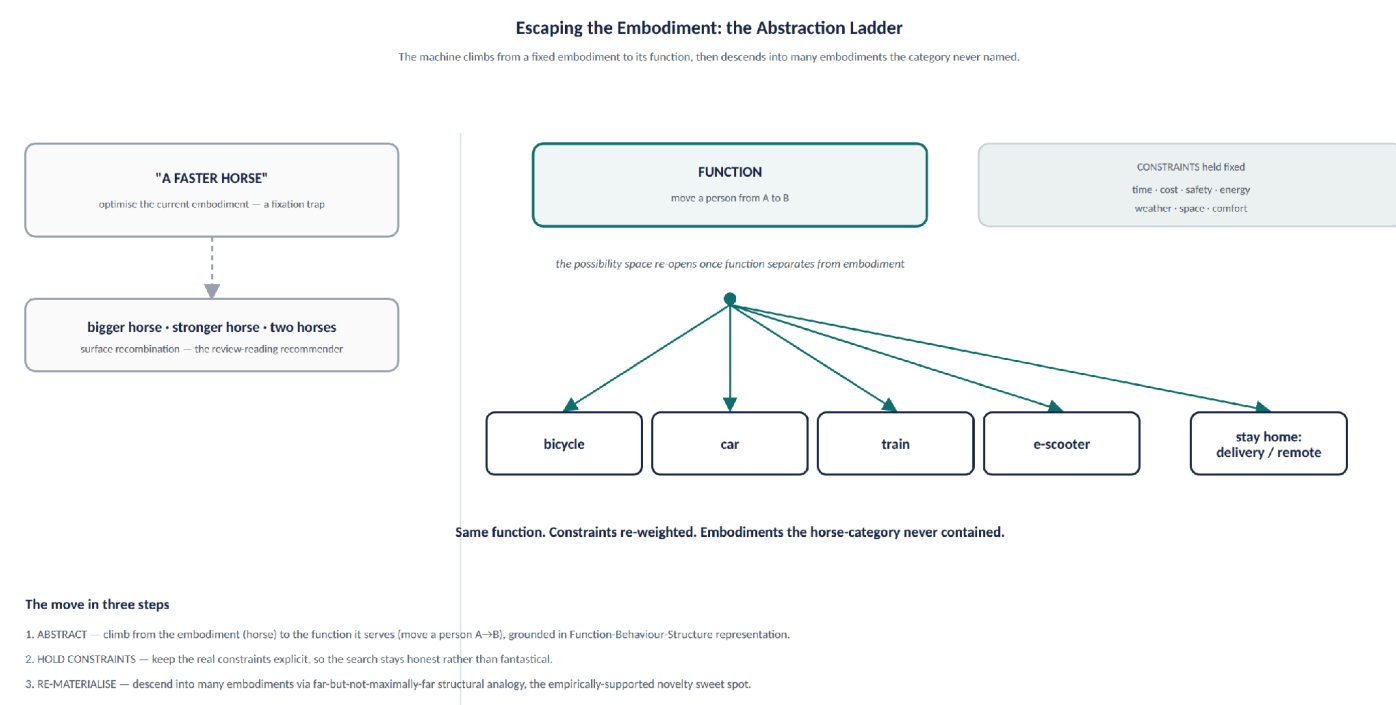


Figure 7. The abstraction ladder — climb from a fixed embodiment to its function, then descend into embodiments the category has yet to name.

12.3 How companies deal with it — and what happens to those that fail

The pattern repeats across industries: separate function from embodiment and survive the shift; optimise the embodiment and get a very good version of an obsolete thing.

Company	Dealt with it — or failed	What happened
Netflix	DEALT — function was 'watch what you want, when you want'	Moved to streaming before the disc business peaked; the function outlived its embodiment
Amazon	DEALT — function was 'buy anything with least friction'	Escaped 'bookstore' into everything, then cloud, devices, logistics
Apple	DEALT — the smartphone abstracted 'phone' into 'a computer you carry'	Redefined the category rather than building a better handset
Kodak	FAILED — held 'film' as the function, though it invented the digital sensor (1975)	Optimised film while digital erased the embodiment; bankrupt 2012
Nokia	FAILED — held 'the best handset' as the function	Superb hardware while the function moved to 'a computer that also calls'; sold its phone business 2014
Blockbuster	FAILED — held 'the rental store' as the function; declined to buy Netflix	Optimised stores and late fees until streaming removed the reason to visit; bankrupt 2010
Xerox	FAILED — held 'the copier' as the function, though PARC invented the GUI, mouse, Ethernet	Let others commercialise the future it invented

What happens to companies that fail the “escape”: they do everything right at the embodiment level — better film, handsets, stores — and lose anyway, because a rival redefined the **function**. Facing fast entrants who iterate the current embodiment faster and cheaper, competing at the embodiment level is the losing

game. The Magic Box suggests to exist so Versuni competes one level up, at the function, where a defensible non-obvious position is still open.

12.4 Versuni as a capability library, and the seven searches

Versuni category	Structural capabilities to extract
Floor Care	Mobility, navigation, mapping, localized action, dust sensing, battery/motor engineering
Coffee	Preference learning, precision, repeatable control, water systems
Kitchen	Airflow, heat, humidity, multi-stage process control, rapid-air technology
Garment Care	Steam generation, material interaction, portable treatment
Climate Care	Environmental sensing, airflow, filtration, environmental intervention
Connected / digital	HomeID user base, telemetry, personalization, connected intelligence

The Magic Box runs seven structured searches, each uniting two spaces the category keeps apart: Function × Capability; Friction × Capability; Path × Capability; Science-mechanism × Human-need; Patent-movement × Product-absence; Competitor-assumption × Counterfactual; Nature-mechanism × Function.

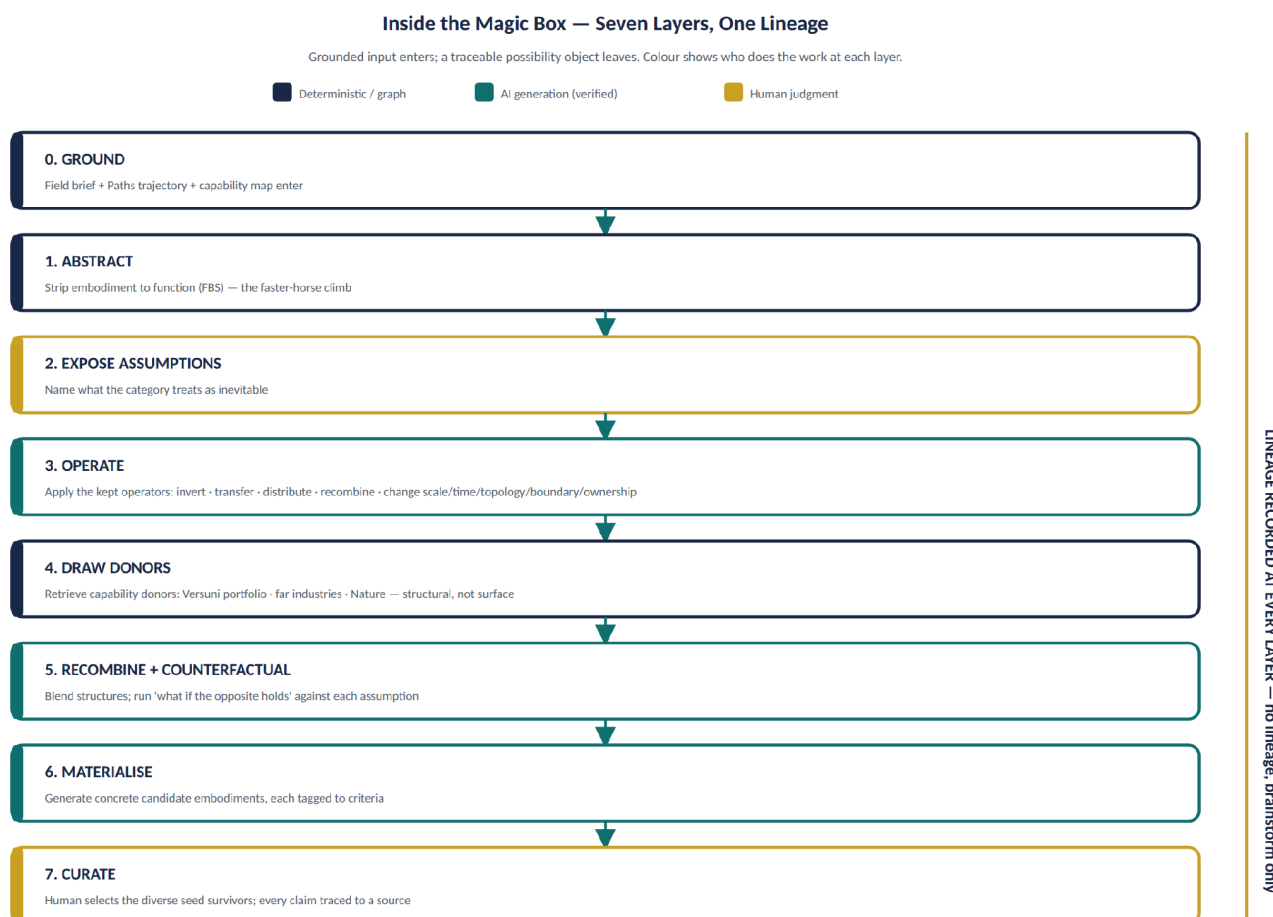


Figure 8. Inside the Magic Box — eight layers, colour-coded by who does the work, with lineage recorded throughout.

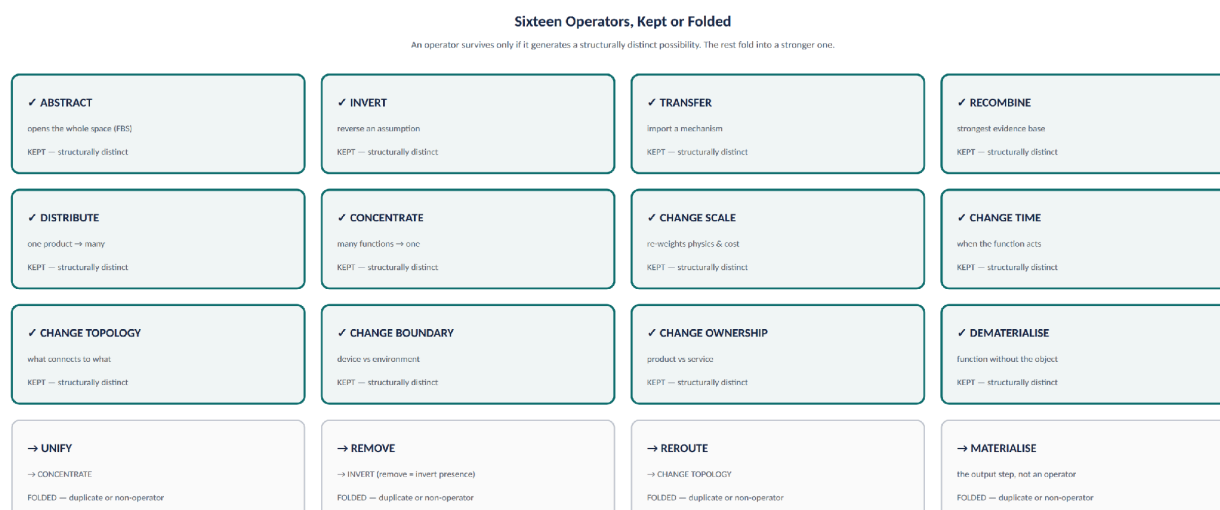


Figure 9. Sixteen candidate operators, kept only where each generates a structurally distinct possibility (twelve kept, four folded).

12.5 Why each operator earns its place

An operator is kept only when it produces a structurally different possibility. Each is stated with what it reveals and when it adds little.

Operator	Why use it	Possibility it can reveal	When it is useless
Remove	Strip a part the category treats as essential	A simpler architecture that a whole part vanishes from	When the part carries the core function
Unify	Fuse two functions into one element	A multi-function device that saves space and cost	When the two functions fight each other physically
Invert	Assert the opposite of a fixed assumption	The box moves to the person, rather than the person to the box	When the assumption is a hard physical law
Transfer	Import a capability from another category or Nature	Floor-care navigation inside an air device	When only a surface analogy holds
Move	Relocate where the function happens	Treatment at the occupied zone, rather than room-average	When location is fixed by physics
Distribute	Spread one central function into many small ones	Distributed sensing across a room	When coordination cost outweighs the gain
Concentrate	Collapse many parts into one strong core	One precise unit replacing several weak ones	When redundancy is the safety feature
Change scale	Shrink or grow the solution	A personal unit, or a whole-home system	When performance breaks at the new size
Change time	Shift when the function acts	Act before air degrades, rather than after	When the trigger must stay reactive
Change boundary	Redraw what is inside the product	The home becomes part of the device	When the boundary protects safety or IP
Change payer / ownership	Shift who pays or owns	Air-as-a-service, rather than a bought box	When customers value ownership itself
Change delivery	Rework how it reaches the home	Flat-pack or installed-service models	When delivery is already the cheap part
Recombine	Merge two architectures into a new one	A hybrid the category has yet to name	When the merge adds parts yet little value

Lineage is mandatory: a serious possibility keeps its full lineage (Appendix D). One that keeps it is a machine output; one that loses it is a brainstorm. The AI proposes and expands, on a human-curated seed (Livotov &

Mas'udah [1]; Feng et al. [14]) that manages the novelty-usefulness trade-off (Kokate & Onkar [8]); every donor, mechanism and figure traces to a retrieved source, keeping the AI clear of inventing evidence.

13. Innovation Evolution, Capability Transfer, and Nature



Innovation evolution — possibilities branch, some die, some strengthen; selection under many reality pressures, rather than a single score.

The machine refuses generate-score-winner and its blended master score. Possibilities evolve as a population tested against nine reality pressures — physics, human behaviour, home/space, economics, competition, technology, manufacturing, regulation, failure. They may die, mutate, merge, split, simplify, or donate a component; rejected learning is preserved rather than lost.

Possibilities Evolve — They Are Not Scored Into a Winner

A population meets nine reality tests. Weak options die; survivors mutate, merge, split, simplify — and donate parts.

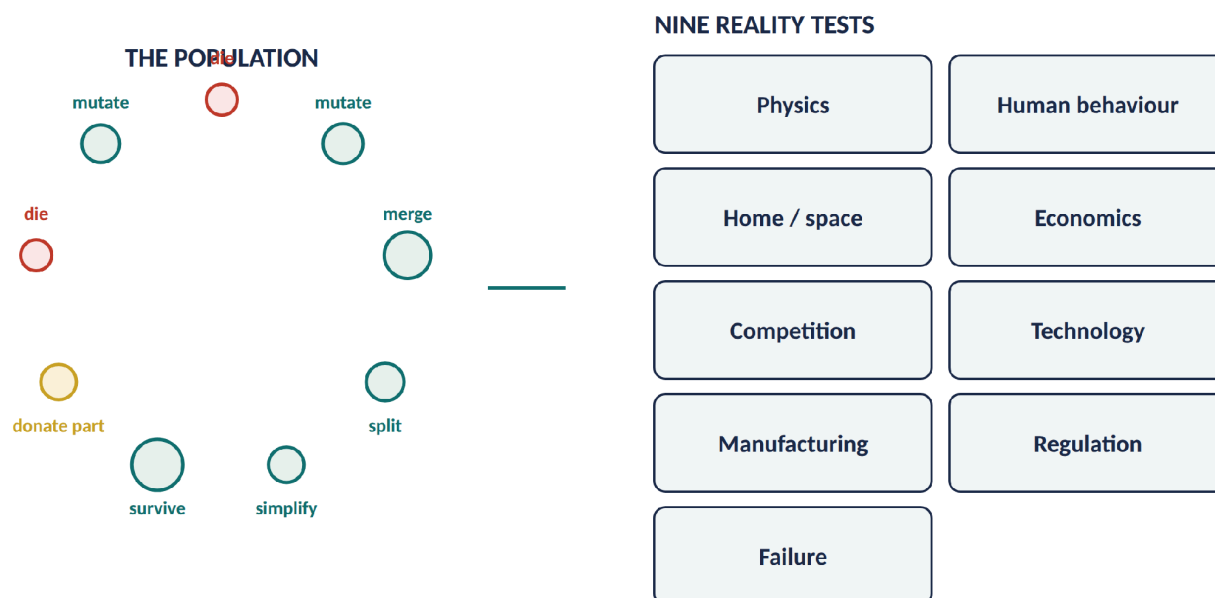


Figure 10. Possibilities evolve — a population meets nine reality tests; weak options die, survivors mutate, merge, split, simplify, and donate parts.

Capability transfer is structural, rather than superficial (far beyond an 'air purifier + coffee machine'). Each transfer — mobility from Floor Care, preference-learning from Coffee, sensing across categories — is a hypothesis carrying its own falsifier, tested in evolution. Nature is a function-first mechanism library (Cui et al. [15]; Chen et al. [16]) returning one of three verdicts — matched, no strong verified analogue, or not applicable; a machine that always finds a biological match is inventing them.

14. The Generative Engine — the Daily Innovation Cycle

The stages so far describe a funnel that could sit still between meetings. This section makes it move. The machine actively creates innovation candidates on a daily cadence, so that idea fabrication becomes a running process rather than an event. On a quiet day it evolves what it holds; on a day reality moves, it fabricates.

The daily innovation cycle: every operating day the machine runs a fixed loop around its idea population. Cheap steps run always; the expensive fabrication steps fire when an opportunity-generating change appears.

1. Load the accepted state and any new evidence since the last run.
2. Identify the changes that generate opportunity — the ones that open, close, or reshape a possibility.
3. Select the Paths those changes touch, and the capabilities in reach.
4. Abstract each affected product to its function, and challenge the assumptions the category holds fixed.
5. Run the Magic Box transformations — invert, transfer, distribute, concentrate, change boundary, scale, time, ownership, dematerialise, recombine.
6. Generate structurally different candidates from those transformations, each with full lineage.
7. Compare against the existing population — merge duplicates, and mutate the promising into variants.
8. Attack every candidate with the criteria, and persist the lineage of survivors and of the killed alike.

9. Surface the three to seven strongest new or changed candidates for human attention.

The Daily Innovation Cycle

Every operating day the machine evolves its idea population — and surfaces 3–7 of the strongest new or changed candidates.

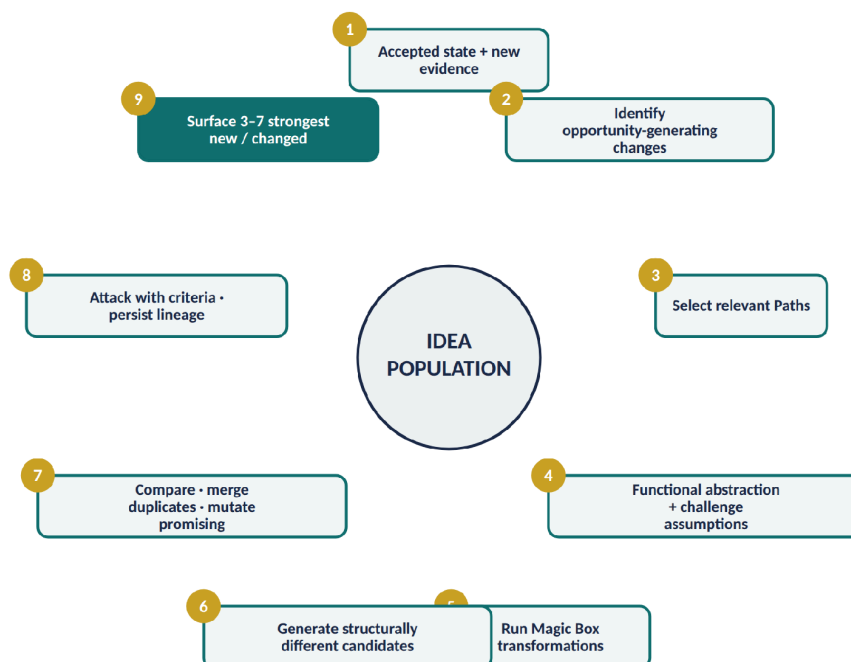


Figure 11. The daily innovation cycle — nine steps around the living idea population; every operating day the machine evolves the population and surfaces three to seven of its strongest new or changed candidates.

14.1 The daily innovation output — three to seven candidates

The visible product of a day is a short, high-value set: three to seven candidates, chosen for strength rather than volume. Each one arrives as a complete, testable object, so a reader can judge it and act on it the same day.

Field on each candidate	What it records
One-line idea	The candidate stated in a single sentence a developer can picture
Source Path	The trajectory in reality it descends from
Assumption challenged	The category assumption it negates or sidesteps
Why now	The change that made it fabricable this week rather than last year
Versuni capability	The owned capability it draws on — the reason Versuni holds an edge
Key features	The few features that carry the value
Economic logic	How it could make money — price, cost, margin, recurring revenue, or saved cost
Competitive difference	What keeps it distinct from what fast entrants already ship
Biggest uncertainty	The single thing most likely to make it wrong
Next experiment	The cheapest test that would move belief the most

14.2 Generative quality — measuring the search, rather than the pile

Idea count alone is a weak KPI (a convergent, deepening strategy can lift adoption more than raw volume, Komura & Yamada [24]): a machine can flood a page with variations of one thing and look productive. The

generative KPIs measure the quality of the search itself — how well-lineaged, how structurally varied, and how survivable the population is. Each names the perverse incentive it guards against.

Generative KPI	What it measures	Healthy direction
Lineage completeness	Share of candidates with a full trace to a source signal	≥ 98%
Structural distinctness	Share of candidates opening a genuinely different architecture	Rising
Assumption novelty	How often candidates challenge an assumption held fixed before	Rising
Versuni leverage	Share of candidates drawing on an owned Versuni capability	High
Economic plausibility	Share with a coherent price-cost-margin story	Rising
Experimentability	Share carrying a cheap, decisive next experiment	High
Survival	Share of surfaced candidates alive after evolution	Honest — high enough to matter, low enough to prove real culling
Learning velocity	Speed at which a candidate moves from idea to a tested belief	Rising
Evidence-to-idea latency	Days from an evidence change to a fabricated candidate	Falling
Rejected-DNA reuse	How often a killed candidate's part strengthens a later one	Rising

The generative test, stated plainly: a strong day produces few candidates, each well-lineaged, structurally distinct, and carrying a cheap decisive experiment — and the population as a whole keeps improving as rejected parts return in stronger form. A weak day produces many near-duplicates with thin lineage. The KPIs make the difference visible.

15. Criteria Throughout the Machine

#	Criterion	Role	Applies mainly at
1	Evidence integrity	GATE	Radar, Paths
2	Human value	GATE	Field, Magic Box
3	Disruption (structural distinctness)	DIAGNOSTIC	Magic Box
4	Versuni advantage	DECISION	Innovations
5	Competitive differentiation	DECISION	Innovations
6	Behaviour / adoption	DIAGNOSTIC	Innovations, New Product
7	Home / space fit	DIAGNOSTIC	Innovations
8	Economic reality	DECISION	Innovations, New Product
9	Physical / technical feasibility	GATE	Innovations, New Product
10	2–5 year execution feasibility	GATE	New Product

Criteria live at every transition, as a ten-level hierarchy split into gates (pass/fail), diagnostics (inform), and decision variables (decide among survivors). **Air Treatment official case:** preserve three decision variables — Consumer Pain, Economic Value, 2–5 Year Feasibility — and compare candidates by Pareto / non-dominated ranking. A candidate leading on all three and sitting on the non-dominated frontier stays visible even when a rival scores higher on one axis alone; the human judges among the frontier, rather than trusting a weighted average.

16. What Versuni Must Investigate Internally

The machine reaches full strength with the people who know reality. Run short structured interviews and one workshop, prioritised by value (full role-by-role questions in Appendix D).

Role (priority)	High-value questions (short)	Artifact	Informs
Disruptive Innovation lead (P0)	Where has innovation stalled? What got killed and why? What is a real win in 12 months?	Prior concepts + kill reasons	All
Product developers (P0)	What is hard today? What almost worked? Which capability is under-used across categories?	Capability + failure notes	Magic Box, Innovations
Consumer insight (P0)	Which complaints recur across categories? Who is a lead user? What do people work around?	Review corpora + personas	Field
Category / product mgmt (P0)	Where is growth, where is it flat? Which adjacencies are open?	Category economics	Innovations, New Product
Engineers + designers (P1)	Real airflow/energy/noise trade? Which motors/filters are sourced? Which form factors fail in the home?	Performance envelopes; design post-mortems	Innovations, Field
Finance + manufacturing + IoT + IP/regulatory (P1)	Margin structure? Consumable economics? What is cheap at scale? What does HomeID data show? Where is IP thin, what regulation is changing?	Cost/pricing models; BOM; telemetry; IP map	Innovations, New Product, Radar
Service + sustainability + sales (P2)	What breaks in the field? What do customers ask for that no product answers?	Failure/returns data	Field, Innovations

Order: P0 (five interviews) in week one, P1 across weeks two and three, P2 in week four, plus one cross-functional workshop in week two to pressure-test the first Paths and seed the first Magic Box.

17. Intelligence Layers — What the Machine Must Know

Everything upstream rests on what the machine actually knows about the world. This section names the intelligence layers, and holds each to the two questions the whole paper turns on: what is reality doing, and what could Versuni create now because of it. Intelligence earns its keep only when it feeds idea fabrication, rather than filling a dashboard.

The first-principle test for every layer — why do we need this: a layer earns its place only when it can change one of six concrete things — a Path, an assumption, a product architecture, an economic model, an experiment, or an innovation decision. Layers are retained on that test alone; a layer that would move a chart yet leave every decision as it was stays out.

Effort follows priority: P0 layers take about 70% of the effort, P1 about 20%, and P2 about 10% — the final tenth reserved for validating and adjusting the machine rather than widening it early. Economics leads, because a disruptive idea that ignores what people can pay and what a product costs stays a poster, rather than a product.

17.1 The twelve intelligence layers

Intelligence · Priority	Why it matters · funnel use	Core questions	Data · quality achievable	Source-API · cadence	Ideas it can trigger
Consumer economics · P0	Sets what households can afford and when they replace; gates every product's price and reshapes Paths and economic models	Disposable income? Expenditure by category? Inflation, energy and housing costs? Affordability? Replacement cycles? Consumables and repair costs? Willingness to pay?	Eurostat / ECB / CBS / OECD series — OBSERVED, official, current	Eurostat SDMX · ECB · CBS OData · OECD · monthly–quarterly	Air-as-a-service where upfront cost blocks adoption; low-consumable designs when running cost bites
Product economics · P0	Sets whether a candidate can make money — the economic gate for every survivor	Retail price? VAT? Margin? BOM and manufacturing cost? Freight, warranty, service? Consumables? TCO at 1 / 3 / 5 years? Gross margin?	OBSERVED at retail / EPREL; ESTIMATED for BOM and margin; INTERNAL-DATA-REQUIRED for true cost	EPREL · retail pages · teardown estimates · internal finance · monthly	Recurring-revenue models; TCO-led positioning when a rival wins on sticker price
Competitor economics · P0/P1	Reveals the economic game each competitor is playing; steers where Versuni can win	Revenue? Margin? R&D spend? Recurring revenue? Installed base? Which game is each competitor playing?	OBSERVED from filings and reports where public; ESTIMATED otherwise	annual reports · market data · filings · quarterly	Attack a margin a volume player struggles to defend; own a niche a scale player treats as too small
Product + feature intelligence · P0	The canonical product universe; feeds the Magic Box its raw material to abstract and challenge	What exists? Claimed specs? Which capability, architecture, assumption and constraint does each product carry?	OBSERVED from pages and manuals — high on specs, medium on internal architecture	product pages · manuals · EPREL · continuous	A capability present in one category and absent in another; a shared assumption one product could break
Manufacturing + supply · P0/P1	Sets what is cheap at scale and what is fragile; shapes feasibility and economics	What is cheap at scale? Where is supply concentrated? Which components carry risk?	ESTIMATED externally; INTERNAL-DATA-REQUIRED for true sourcing	trade data · supplier maps · internal · quarterly	Designs that reuse a component Versuni already buys at scale

Intelligence · Priority	Why it matters · funnel use	Core questions	Data · quality achievable	Source-API · cadence	Ideas it can trigger
Tax + policy · P1	Shifts affordability and demand through subsidy, tariff and levy	Which subsidies, tariffs and levies apply? What is changing?	OBSERVED from official sources — high	EUR-Lex · customs · national policy · as-published	Products that qualify for an energy subsidy; timing a launch to a levy change
Technology · P0	Shows which capabilities became feasible or cheap; opens fabrication	What became feasible or cheap? Which applied-AI and sensing methods are maturing?	OBSERVED from code and papers — high on momentum	GitHub · Hugging Face · OpenAlex · weekly	A sensing method now cheap enough to distribute through a room
Patent intelligence · P0/P1	A strong direction signal and a soft commercialisation predictor; read as families	Which applicants, classes and families are growing? Where is Versuni thin?	OBSERVED — use FAMILIES rather than raw counts to keep one invention counted once	Google Patents BigQuery · EPO OPS · monthly	A patent movement pointing at a product still absent from shelves
Human + behavioural · P0	Grounds whether people would adopt — the human-value gate	Which frictions recur? Who is a lead user? What do people work around?	OBSERVED from reviews and field — medium-high with coding	review corpora · field observation · continuous	A workaround so common it names an unmet function
Trends + Paths · P0	The named trajectories the whole machine acts on	From what state to what state? What drives, blocks and reverses it? What falsifies it?	SYNTHESISED from observations; quality set by corroboration	internal Paths sheet · every cycle	Every fabricated candidate descends from a Path
Science + physics · P0	Sets the hard limits and the mechanisms available to transfer	What does physics permit? Which mechanisms are available? Which research fronts accelerate?	OBSERVED from literature — high	OpenAlex · physics constants · continuous	A mechanism from another field that achieves a function more directly
Regulation + standards · P1	Sets what a product must meet and when the bar moves	Which ecodesign, air and safety rules apply? What changes, and by when?	OBSERVED from EUR-Lex — high	EUR-Lex CELLAR SPARQL · standards bodies · as-published	A design that meets a tightening standard a rival struggles to reach in time

17.2 Reading economics honestly — the three-label rule

Economic intelligence is the layer most easily faked, because a confident number hides its own uncertainty. Every economic figure the machine stores carries one of three labels, so a reader knows the confidence behind it before acting.

Label	Meaning	Example
OBSERVED	Read directly from a public source	Retail price from a listing; energy class from EPREL
ESTIMATED	Inferred from teardown, benchmarks, or comparable products	BOM and gross margin from a component estimate
INTERNAL-DATA-REQUIRED	Settled only by Versuni's own records	True landed cost, warranty rate, consumable margin

Why the labels matter for fabrication: a candidate whose economics rest on OBSERVED figures is ready for a real decision; one resting on ESTIMATED figures is ready for an experiment that would confirm them; one resting on INTERNAL-DATA-REQUIRED figures names exactly the internal number the team should fetch first. The label turns a soft guess into a precise next step.

17.3 More families the machine reads

Beyond the twelve, four families sharpen feasibility and delivery, and one family carries the strongest evidence of all.

Family	What it adds	Priority
Logistics + delivery	Packaging, freight, warehousing, last-mile, returns, service — and delivery itself as an innovation variable	P1
Environment + home	Housing, room size, climate, energy, and how the home shapes use	P1
Nature + cross-industry	Structural mechanisms to transfer — distributed sensing, feedback, local response, redundancy	P2
Real experiments	The result of a real test — the highest-quality evidence about a new idea; prototyping tracks technology-readiness progression (Rehberg et al. [20])	P0

17.4 Data quality — what the machine can know, and what needs an experiment

Honesty is a feature: every variable carries a quality class, so a confident-looking number states its own confidence. A candidate is only ever as strong as the class of the evidence under it.

Class	Meaning	Typically here	Example
A — strong	Direct, official, traceable, current	Products, features, prices, tax rates, regulation, papers, patent metadata	Retail price; energy class from EPREL
B — good proxy	Useful with a stated limitation	Review-based frictions; trend direction	Recurring complaint themes across reviews
C — estimate	Requires a sensitivity check	BOM; competitor economics; TCO	Estimated gross margin from a teardown
D — unknown	Requires internal data or an experiment	True WTP; future demand; competitor roadmap; new-architecture performance	Whether households would pay for air-as-a-service

So what for Versuni: the hardest questions — willingness to pay, future demand, and how a new architecture performs — sit in class D by nature, so the machine treats them as experiments to run, rather than numbers to assert.

18. From Data to Ideas — the Transformation Chain

Why this matters: data on its own creates little value. Value appears only when the machine transforms a raw source, step by step, into a testable product idea. This section shows the exact chain, so the path from a number to an experiment is visible end to end.

Source → Observation → Change → Path → Opportunity → Magic Box → Idea → Architecture → Experiment. Each link has an input, a transformation, an output, and a criterion that must pass before the next link may be written.

Link	Input → transformation → output	Criterion to pass
Source → Observation	A raw source value is recorded with provenance	Source quality; a content hash for identity
Observation → Change	A new value is compared with the accepted state	The difference clears a materiality threshold
Change → Path	Related changes are named as a directional trajectory	Independent corroboration; a stated falsifier
Path → Opportunity	A Path is read for what it opens for Versuni	Grounded human value; a live decision it touches
Opportunity → Magic Box	The opportunity enters functional abstraction	The current embodiment is removed; assumptions exposed
Magic Box → Idea	Operators fabricate structurally different candidates	Full lineage; a distinct operator behind each
Idea → Architecture	A candidate is shaped into a concrete product architecture	A coherent price, cost, and capability story
Architecture → Experiment	The riskiest assumption becomes a cheap test	A metric, an expected result, and a kill criterion

18.1 Three concrete Versuni chains

- Air treatment: rising energy prices (Observation, class A) → a Path toward running-cost sensitivity → opportunity in air quality at low running cost → the Magic Box moves treatment to the occupied zone → a mobile personal-air unit → priced against a bought box, tested by a willingness-to-pay experiment (class D).
- Coffee → across categories: precise dosing and preference-learning (Versuni capability) → transferred by analogy into another category → an appliance that learns a household routine → tested by a small in-home trial.
- Floor care → air: mobility, navigation, and mapping (Versuni capability) → transferred into air treatment → a unit that follows the occupied zone → tested against a fixed-box baseline on measured air quality where people actually are.

So what for Versuni: the chain is the machine in miniature — a reviewer can trace any idea back to the source that triggered it, and forward to the experiment that would prove it.

19. Data, APIs, and Simple/Deep Mode

19.1 What Month 1 needs — P0, P1, P2

Domain · source	API / method	Establishes	Stage · Priority
Science — OpenAlex	REST (key from Feb 2026), free	Emerging research fronts, citation acceleration	Radar · P0
Consumer — review corpora (Amazon Reviews 2023)	Bulk / sampled fetch within terms	Recurring frictions, satisfaction drivers	Field · P0
Products — Versuni + competitor pages, manuals	Fetch within terms	What exists, claimed specs, feature gaps	Radar/Field · P0
Versuni capabilities — internal map	Interviews + patent read	What Versuni can already do	Magic Box · P0
Patents — Google Patents (BigQuery) + EPO OPS	SQL / REST, free tiers	New applicants, class growth, citation acceleration	Radar/Magic Box · P0
Economics + prices — Eurostat/ECB/CBS + EPREL	SDMX/OData/REST, free	Household economics, energy prices, energy class	Innovations · P1
Regulation — EUR-Lex CELLAR SPARQL	SPARQL, free	New EU ecodesign / air rules at source	Radar · P1
Technology — GitHub + Hugging Face	REST, free tokens	Applied-AI and open-source momentum	Radar · P1
Environment — OpenAQ + Copernicus CAMS + EEA	REST + cdsapi, free	Air-quality drivers of demand	Radar · P1
Nature — GBIF + EOL (function search via LLM+KG)	REST, free	Biological mechanisms (function-first)	Magic Box · P2
News — GDELT	REST + BigQuery, free	Launch and narrative signals	Radar · P2

P0 = required for the first working machine (OpenAlex, review corpora, product pages, the internal capability map, patents). Building the P0 five is enough for the week-two vertical slice. P1 = high-value next; P2 = Deep Mode.

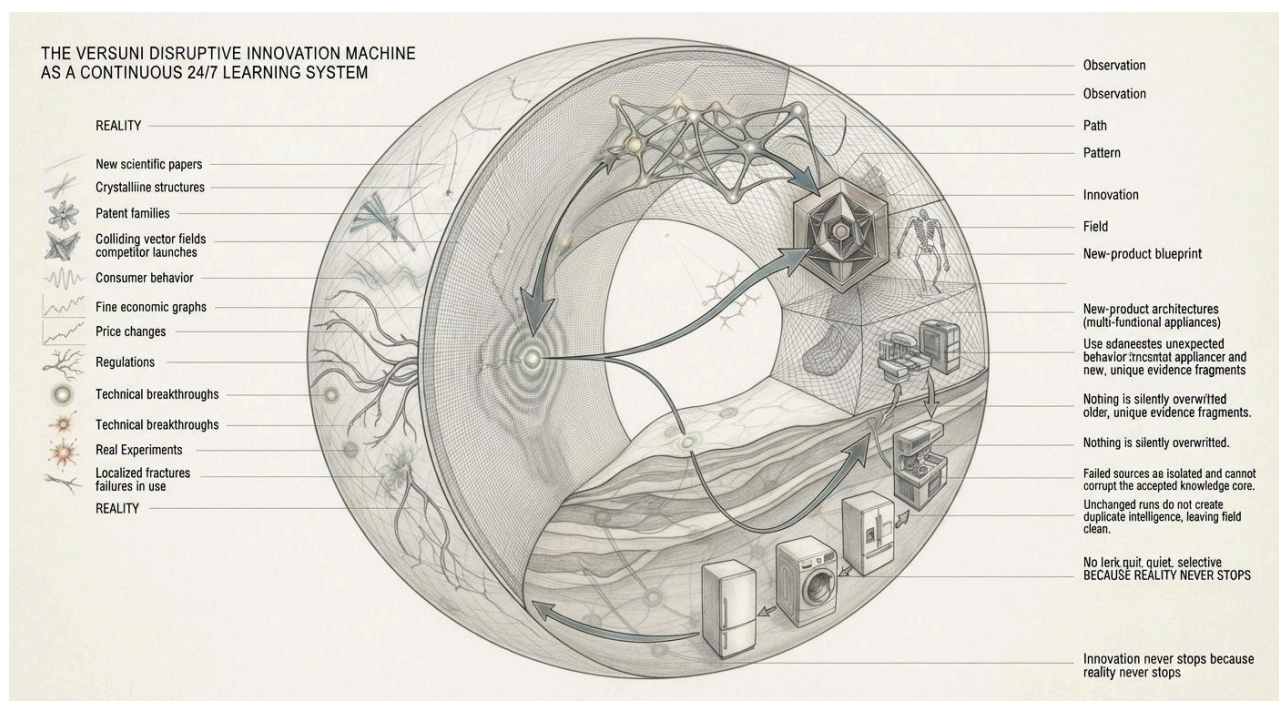
19.2 Simple Mode — exact Excel/CSV sheets

Sheet	Key fields	Links to
SOURCES	source_id, name, authority, domain, access, terms, update_freq, last_pull, health	—
OBSERVATIONS	obs_id, source_id, content_hash, what, where, since_when, value, confidence	SOURCES
PATHS	path_id, from, to, drivers, blockers, time, reversibility, falsifier, evidence_ids, contradiction, robustness	OBSERVATIONS
FIELD	brief_id, path_id, friction, whats_true_now, biggest_uncertainty, what_makes_it_wrong, evidence	PATHS
CAPABILITIES	cap_id, category, capability, strength, patents, breaks_when	—
POSSIBILITIES	poss_id, function, embodiment, operator, assumption_challenged, donor_cap_ids, analogy, nature_relation, falsifier, lineage	FIELD, CAPABILITIES
INNOVATIONS	innov_id, poss_id, state, crit_1..crit_10, kill_reason, parent_ids	POSSIBILITIES
PRODUCTS	prod_id, innov_id, audience, changed_problem, architecture, economics, first_prototype, kill_criterion, key_assumption	INNOVATIONS
RUNS	run_id, timestamp, commit, input_hashes, output_hashes, source_health	(all)

Sheet	Key fields	Links to
KPIS	kpi_id, family, value, target, cadence, owner, status	RUNS

Deeper infrastructure is earned: DuckDB + Parquet when runs outgrow Excel; PostgreSQL on a second concurrent writer; a graph database when capability-transfer search across many nodes becomes the bottleneck; simulation when a prize estimate needs resource-contention or full-posterior modelling.

20. The 24/7 Technical Architecture



The machine as a continuous 24/7 learning system — a loop with no natural endpoint, because reality has none.

24/7 means cheap, deterministic perception running constantly, with expensive reasoning firing only on material change. The perception chain — source change, validate, diff against accepted state, identify affected Paths, check the Field — runs every tick and touches no AI. Only a material change reopens the relevant Magic Box or innovation, versions the state, and reports if warranted.

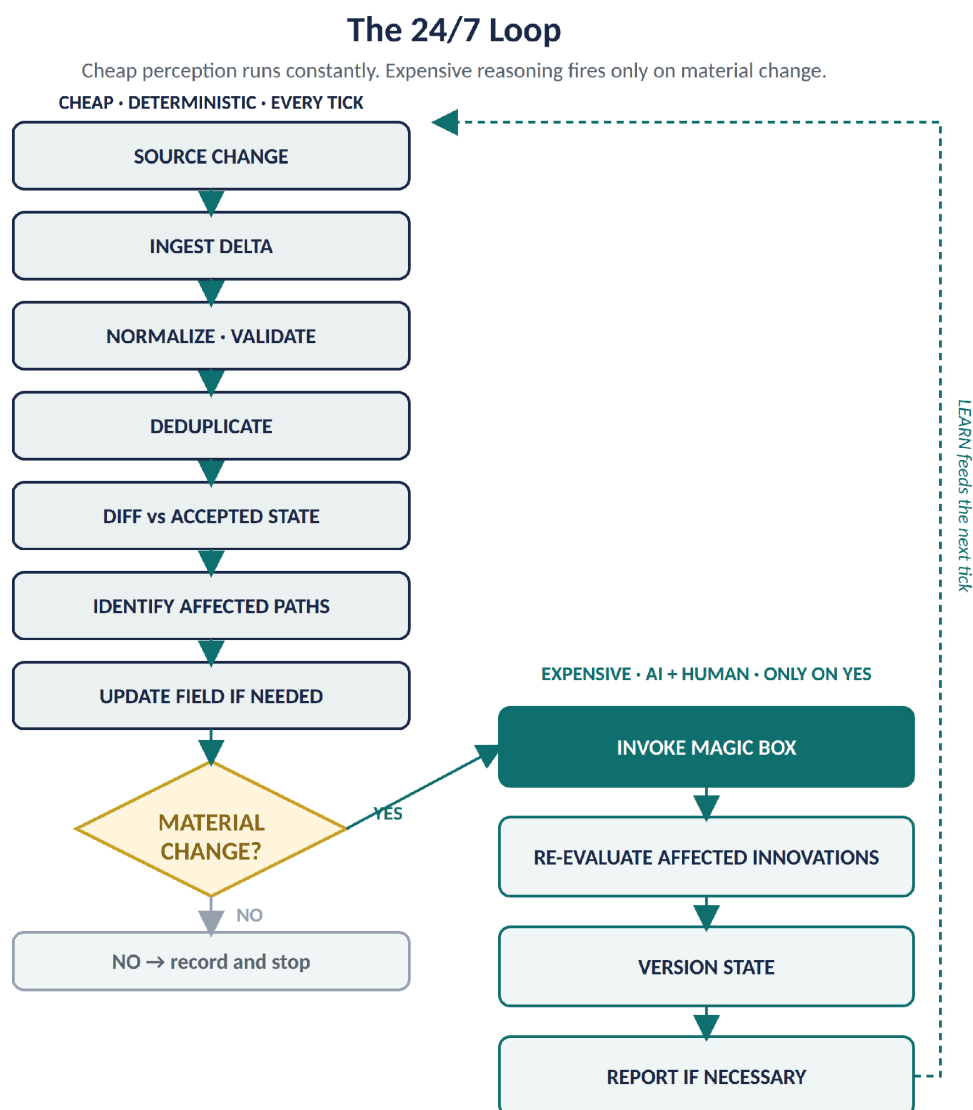


Figure 12. The 24/7 loop — cheap deterministic perception every tick; expensive reasoning only when the material-change gate returns YES.

The Simple-Mode stack (current for August 2026): Python 3.13, uv for reproducible installs, DuckDB + Parquet + Polars as one embedded engine, hand-rolled CUSUM/change-point, sentence-transformers + scikit-learn for clustering, NetworkX on edge tables, NumPy Monte Carlo, GitHub Actions cron, static reports. Reliability rests on idempotency by content hash, HTTP conditional requests, append-only raw with versioned accepted state, retry with backoff, failure isolation through staging, and one run manifest per run. GitHub guardrails keep an AI agent clear of hardcoding truth, rewriting accepted evidence, deleting snapshots, or pushing broken work.

Simple Stack Now • Deep Stack on Trigger

Month one ships the left column. The right column is added only when a named trigger fires.

SIMPLE STACK — build now	DEEP STACK — add on trigger
Python 3.13.15 pinned — the safe production baseline	PostgreSQL a second concurrent writer appears
uv + pyproject + uv.lock fast, hashed, reproducible installs	ruptures hand-rolled change-points fall short
DuckDB + Parquet + Polars one embedded engine, no server	HDBSCAN • UMAP • BERTopic simple clustering visibly fails
NumPy • SciPy • statsmodels stats + hand-rolled CUSUM / BOCPD	rustworkx NetworkX becomes the bottleneck
sentence-transformers + scikit-learn embeddings + simple clustering	FastAPI + Pydantic v2 a real interactive consumer appears
NetworkX on DuckDB edge tables paths graph, rebuilt each run	SimPy • Mesa • PyMC • PyTorch a real world-model need appears
NumPy Monte Carlo directional prize estimates	Prefect a genuine multi-step DAG appears
GitHub Actions cron off-peak, idempotent, catch-up	Temporal durable crash-resume execution needed
Static report artifacts no backend yet	Playwright a real web UI to visually test
pytest + Hypothesis + Ruff contracts, idempotency, lint	Prometheus + Grafana long-running services to watch live
structlog + per-run manifests full traceability, no metrics stack	

Figure 13. Simple Stack now (left); Deep Stack added only when a named trigger fires (right).

21. KPIs and Reporting

Three families, each KPI with a formula, target, cadence, owner and action-when-red (full register in Appendix F). None auto-selects a winning product.

KPI (family)	Formula	Target • cadence	Action when red
Idempotency (Health)	New objects from re-running unchanged input	0 • every run	Freeze features; fix content-hash identity
Provenance completeness (Health)	Share of objects with full lineage	≥ 98% • weekly	Block promotion of any object missing lineage
Independent corroboration (Intelligence)	Share of Paths on ≥ 2 unrelated families	≥ 80% • weekly	Downgrade single-source Paths to watch pool
Evidence-ablation survival (Intelligence)	Share surviving removal of strongest source	≥ 70% • weekly	Rebuild the Path on broader evidence
Time signal→tested possibility (Performance)	Median days change → first experiment	Falling • monthly	Clear the slowest handoff
Time-to-kill (Performance)	Median days candidate → documented kill	Short + honest • monthly	Give the kill authority real teeth

Structurally distinct possibilities (Performance)	Distinct-operator survivors per cycle	Rising · per cycle	Widen the Magic Box seed and operators
Evidence→re-evaluation latency (Performance)	Days evidence change → concept re-score	Falling · monthly	Automate the diff-to-reopen trigger
Report	Trigger	Content · action	
Material alert (real-time)	A Path clears corroboration + contradiction and touches a live decision	One event + its evidence · decide	
Daily pulse	Scheduled — 'nothing advanced' honestly when quiet	What moved, what stalled · scan	
Weekly report	Scheduled	Paths, possibilities generated/killed, experiments · review	
Monthly disruption review	Scheduled	The few Paths with genuine disruptive potential · steer	

-message logic: silence is the default on a quiet day; the bar rises with the volume of the channel — a monthly escalation keeps its force, a daily one becomes noise.

Generative KPIs sit alongside these: the ten measures of search quality — lineage completeness, structural distinctness, assumption novelty, Versuni leverage, economic plausibility, experimentability, survival, learning velocity, evidence-to-idea latency, and rejected-DNA reuse — live in the Generative Engine section (the daily innovation cycle), because they measure the fabrication engine rather than the machine's health or its intelligence.

22. Team and Governance

The machine is built and run by the Disruptive Innovation team — one accountable owner with data engineers, designers and consulted specialists. Governance is organised around clear ownership questions, each with exactly one owner: consumer insight and science own evidence; innovation leadership and AI engineering own the Magic Box; innovation leadership holds the single named authority to kill; engineering and simulation own technical validity; economics owns economic validity; innovation leadership approves movement to New Product; and human judgment stays load-bearing at the four points machines handle worst (co-creation gains shift with expertise, Wang et al. [25]; human-plus-AI judges novelty best, Wu et al. [26]) — exposing an unspoken assumption, curating a diverse seed set, judging whether a transfer is genuine, and deciding to kill.

23. How to Build It — and the Claude Code Plan

23.1 Claude Chat and Claude Code — who does what

Two tools, two jobs. Chat does the thinking work; Code does the building work. Keeping them separate keeps each prompt small and each result checkable.

Claude Chat — reasoning	Claude Code — building
Research and paper synthesis	Data schemas and connectors
Counterfactuals and Magic Box reasoning	Normalisation, calculations, the Path pipeline

Claude Chat — reasoning	Claude Code — building
Critique and hostile review	The Magic Box pipeline, lineage, and tests
Naming Paths and curating the seed set	The scheduler, KPIs, reports, and the interface

23.2 Elements to have now, and what waits

Must-have in Month 1	Waits for a trigger
Python 3.13 + uv lockfile; Excel/CSV sheets; the P0 five sources	Python 3.14; DuckDB/Parquet; P1/P2 sources
Content-hash idempotency + append-only raw + run manifests	PostgreSQL (second concurrent writer)
Hand-rolled change-point + simple clustering; NetworkX edge tables	ruptures/HDBSCAN/BERTopic; a graph database
NumPy Monte Carlo; GitHub Actions cron + protected main + CI	SimPy/PyMC/PyTorch; Prefect/Temporal
Static reports + the ten sheets; the Magic Box + vertical slice	FastAPI backend; a polished UI

23.3 The one vertical slice — build this before anything wide

The slice: Real Source → Observation → Path → Field → Magic Box → Innovation → Product Hypothesis → Experiment → trace back to source. One of each, full lineage. Why first: it proves the whole machine end to end; a thin whole beats a thick fragment.

23.4 Order of priorities — dependency-driven

- **FIRST:** the ten-sheet data model + one P0 source with idempotent ingestion — everything depends on it.
- **SECOND:** the complete thin vertical slice with full lineage — the week-two proof.
- **THIRD:** deepen perception (more P0/P1 sources, corroboration gates) and the Magic Box operators + capability search.
- **FOURTH:** the evolution/kill mechanism, then KPIs, reporting, the 24/7 loop, and the robustness tests.
- **WAITS:** any backend, orchestration engine, graph database, polished UI, and all P2 sources.

23.5 The eight Claude Code tasks

How many prompts: eight bounded tasks, each one task, one outcome, one review, one commit. How much time: ~3–5 focused hours per task across the 30 days. The repository is durable memory: CLAUDE.md holds the stack, the ten-sheet contract, the lineage rule and guardrails; STATE holds the live cycle; REQUIREMENTS and DECISIONS hold the spec and every choice; the task prompt holds one task; just-in-time context is pulled by tool call.

Four Tiers of Context — Never One Giant Prompt

Each tier maps to a real Claude Code mechanism. Smaller, high-signal context beats a bloated permanent prompt.

ALWAYS IN CONTEXT

Stable project rules, read every session

→ *CLAUDE.md* — kept short, or it gets ignored

JUST-IN-TIME

Pulled at runtime by reference

→ *Skills* · *@file* · *gh* / *CLI* / *MCP tool calls*

TASK-SPECIFIC

Loaded for one task, then dropped

→ *Subagent context windows* · *per-task SPEC.md* · */clear*

REFERENCE-ONLY

In the repo, read on demand

→ *Grep* / *Glob* / *Read* · *git history* · *linked docs*

Figure 14. Four tiers of context — what stays in *CLAUDE.md*, what loads just-in-time, per task, or on demand.

Task	Exact output	Pass condition
Task 0 — Repo + requirements audit	Scaffolded repo: <i>CLAUDE.md</i> , <i>REQUIREMENTS</i> , <i>DECISIONS</i> , <i>STATE</i> , empty sheets, uv lock, CI	CI green on an empty test; the lineage rule exists
Task 1 — Data model + Excel contract	Pydantic models for all ten objects; CSV read/write; a data-contract test	A sample row round-trips; a no-lineage possibility is rejected
Task 2 — Radar + ingestion	Idempotent ingestion (OpenAlex) with <i>content_hash</i> + run manifest + <i>ETag/cursor</i>	Re-running unchanged input creates zero new rows
Task 3 — Observations → patterns → paths	Clustering + hand-rolled change-point; <i>PATHS</i> with nine fields + corroboration/contradiction	A seeded Path carries all fields and survives strongest-source removal
Task 4 — Field + world model	<i>FIELD</i> briefs grounding each Path; thematic coding; uncertainty fields	Each live Path yields ≥ 1 grounded brief
Task 5 — Magic Box + capability transfer	The seven searches; twelve operators; <i>POSSIBILITIES</i> with full lineage; structural analogy	Every possibility traces to a source and a distinct operator
Task 6 — Innovation gates + evolution	Unblended criteria scoring; documented kills; survivor population; rejected-DNA store	A killed candidate carries a reason; a no-change run kills nothing new; Pareto view for Air
Task 7 — KPIs + reporting + 24/7 + trace + final E2E audit	Three KPI families; four reports; GitHub Actions loop; trace view; full test suite	One evidence item travels source→product with lineage; changing it changes the output; all tests green

What the Agent Cannot Do — and What Stops It

Each risk is blocked by a concrete GitHub mechanism, not by trust.

FORBIDDEN ACTION	GITHUB MECHANISM THAT BLOCKS IT
⊗ Push broken work	✓ Required CI status checks before merge
⊗ Declare visual work done untested	✓ Required Playwright / screenshot check
⊗ Rewrite accepted evidence	✓ CODEOWNERS review on /accepted/**
⊗ Change decision criteria silently	✓ CODEOWNERS review on /criteria/**
⊗ Delete historical snapshots	✓ Block force-push + restrict deletions
⊗ Self-approve its own work	✓ Require review from someone other than author
⊗ Fill in human validation	✓ Protected human-only sign-off paths
⊗ Touch production secrets	✓ Environment protection + required reviewer

Figure 15. GitHub guardrails — each thing the agent must be kept from doing, blocked by a concrete mechanism.

The literal prompt shape per task: "Read CLAUDE.md, REQUIREMENTS.md and STATE.md. Implement <Task N objective>. Write to <files>. It passes when <pass condition>. Add tests that prove it. Show me the test output and a diff, then stop for review before commit."

24. The 30-Day Chronogram

The month has four purposes — See, Understand, Invent, Operate — and the effort is weighted toward invention. The build starts with the innovation capability, rather than a website.

Week · purpose	Share	Build	Output · PASS
Week 1 — See	25%	Versuni capability map, consumers, competitors, economics, science, technology, source and assumption registries	A credible reality base
Week 2 — Understand	25%	Change detection, patents, trends where justified, Paths, category assumptions, economic and technology movement	A small set of defensible Paths

Week · purpose	Share	Build	Output · PASS
Week 3 — Invent	30%	Magic Box, capability transfer, counterfactuals, idea population, criteria, economics per idea, first experiments	Several structurally different, traceable product hypotheses
Week 4 — Operate	20%	Automation, the daily cycle, KPIs, reports, lineage, experiments, end-to-end test	A working continuous process: source → Path → idea → hypothesis → experiment → learning

A full thin machine runs by the end of week two; weeks three and four deepen and harden it. The spine is dependency-driven, so the vertical slice stays unblocked.

Days	Objective (task)	PASS criteria	If it fails
1–3	Scaffold + data model (Task 0–1)	A sample row round-trips; a no-lineage possibility is rejected	Fix the contract before adding a source
4–7	Radar + one PO source (Task 2); start PO interviews	Re-running unchanged input creates zero new rows	Fix content-hash identity first
8–14	Paths + Field + Magic Box + one Innovation (Tasks 3–5, thin) — the vertical slice	A reviewer traces the hypothesis to its source; changing the evidence changes the output	Stop everything; a slice without lineage is the project failing quietly
15–21	Deepen perception + generation (Tasks 3–5 wide, Task 6)	A Path survives corroboration + contradiction; ≥ 1 possibility killed with a reason	A week with zero kills means the kill authority needs teeth
22–30	Harden + report + prove (Task 7)	Every definition-of-working test passes, above all: changing real evidence changes the reasoning	A failing test is a real defect; fix before Day 30

25. Definition of Working

The machine passes only when changing real evidence changes its reasoning. The suite tests that, plus the failure cases around it:

- Lineage travel — one evidence item travels source → observation → path → field → possibility → innovation → product hypothesis → experiment, with complete backward lineage.
- No-change run — re-running unchanged input creates zero new intelligence objects.
- Contradictory evidence — evidence that contradicts a live Path visibly weakens or kills it.
- Source failure — one failed source stays isolated in staging and leaves accepted state intact.
- Path/innovation death — a Path or possibility that fails its falsifier dies, with its learning preserved.
- Economic + behavioural sensitivity — changing a prize or behaviour assumption changes the ranking of survivors, provably.
- Historical reproduction — a stored run reproduces from its manifest exactly.
- Zero hardcoding — changing the data changes the output; the GitHub guardrails keep hardcoding out of reach.

26. What Can Go Wrong — the Failure Audit

Why this matters: a machine that acts on evidence can be fooled by bad evidence. This audit names how each part can fail, how the machine catches it, prevents it, and recovers — so a failure becomes a handled event, rather than a silent one.

Failure	Why it matters	Detection	Prevention	Recovery	Owner	
Bad data	Wrong, stale, duplicated, or context-poor data	Every idea inherits the flaw	Freshness checks; duplicate hashing; source-health flags	Content-hash identity; corroboration across families	Roll back to the last accepted snapshot	D a t a e n g i n e e r i n g
Bad research	Weak paper; wrong population; correlation read as cause	A single study steers a design	Method and population fields on every source	Two-family corroboration; verdict downgrades	Reweight the design; mark the claim provisional	F o r e s i g h t
Bad trend	One observation dressed as movement; short-term noise	A Path forms on sand	Temporal-evidence and persistence fields	Require time-ordered evidence and a falsifier	Retire the Path; keep the observation	F o r e s i g h t
Bad patent read	A filing read as a product; abandoned claims counted	A direction signal turns into a false promise	Legal-status and family fields	Work by family; check status; treat as direction	Re-label as a soft commercialisation signal	I P w a t c h
Bad economics	Price read as willingness-to-pay ; missing VAT, channel, geography	The business case misleads	The three-label rule (Observed / Estimated / Internal)	Separate price from WTP; complete the TCO	Mark the figure class D; schedule an experiment	F i n a n c e

Failure	Why it matters	Detection	Prevention	Recovery	Owner	
Bad competitor intel	A marketing claim read as capability; a gap read as opportunity	Strategy bets on a mirage	Claim-versus-evidence tagging	Ground claims in teardown or review evidence	Downgrade to a hypothesis	Categorization management
Bad Magic Box	Random brainstorming; semantic variants; impossible physics; weak analogy	Pseudo-novelty crowds out real novelty	Structural-distinctness and lineage checks	Function-first abstraction; structural-analogy test; physics gate	Fold weak variants; keep the structural core	Innovation attribution lead
Bad AI	Hallucination; invented sources; false confidence; prompt injection	The machine trusts a fiction	Every claim carries source, date, epistemic state	Least-privilege access; retrieval-grounded output; injection tests	Quarantine the output; human review	Alignment engineering
Bad human decision	Executive bias; politics; favourite-project protection; premature killing	A strong idea dies or a weak one survives for a wrong reason	Documented kill reasons; a Pareto view	Criteria-gates; one named kill authority; recorded rationale	Reopen from lineage; revisit the record	Innovation attribution lead

Failure	Why it matters	Detection	Prevention	Recovery	Owner	
Bad automation	Duplicate runs; API failure; silent overwrite; stale state; broken scheduler	The record drifts from reality	Run manifests; idempotency checks; scheduler health	Content-hash idempotency; append-only raw; retries with backoff	Replay from the manifest	D a t a e n g i n e e r i n g
Bad innovation KPI	Rewarding idea count, activity, or novelty alone	The machine games the metric	KPI review against perverse incentives	Measure lineage, distinctness, survival, and learning	Retire the metric; keep the outcome	I n n o v a t i o n l e a d
Bad security	Unauthorised data; secret leakage; uncontrolled writes; external action	Trust and compliance break	Audit logs; permission and secret scans	Role-based access; secrets management; approval before external action	Revoke access; rotate secrets; review the log	A d m i n + A l e n g i n e e r i n g

So what for Versuni: every failure family has a named detection, a control, and a recovery — so the machine stays trustworthy on a bad day, which is the day that matters.

27. The Versuni AI Constitution

Why this matters: the constitution governs every AI interaction with the machine. It stays short enough to operate, and it keeps the machine an assistant to human decision, rather than a decision-maker.

Priority	The rule
1 — Reality first	AI works from evidence. Every factual claim carries a source, a date, context, and an epistemic state. AI may create hypotheses, counterfactuals, ideas, and interpretations, kept clearly apart from evidence.
2 — Human authority	Humans hold accountable authority over strategy, validation, major economic assumptions, engineering sign-off, legal and regulatory judgment, experiment approval, investment, and final decisions. AI assists the decision.
3 — Traceability	Every important output keeps its lineage — Source → Observation → Path → Magic Box → Idea → Experiment — so any reader can trace why it exists.
4 — Permission before action	Least-privilege access, with separate rights to read, analyse, suggest, create a candidate, write internal state, run a test, send an external message, change decision state, and approve. The default AI role stays within read, analyse, suggest, create-candidate, and safe tests.
5 — Uncertainty is visible	Explicit states — Observed, Derived, Inferred, Hypothesis, Simulated, Human judgment, Unknown. Confidence follows evidence quality, and Unknown stays Unknown.
6 — Fail safe	When evidence, APIs, permissions, or state look unreliable, the machine preserves the last accepted state, records the failure, surfaces it, limits downstream spread, and requests human review. Reliability outranks apparent completeness.

27.1 What AI does, and what needs human authority

AI can do this autonomously	This needs explicit human authority
Research public sources; summarise papers; extract structured evidence	Change accepted evidence, or overwrite an accepted snapshot
Compare products; identify contradictions; detect changes	Change the final evaluation criteria
Propose Patterns and Paths; generate counterfactuals	Create a human-validation label
Run Magic Box transformations; suggest product possibilities	Approve engineering feasibility, health, safety, or legal conclusions
Challenge assumptions; suggest experiments; create simulations	Select the final investment, or launch a product
Explain lineage; prepare internal reports	Send a consequential external communication
Generate code; run approved automated tests	Change access permissions, or delete historical evidence
Surface uncertainty and open failures	Present a hypothesis as observed reality, or invent a source, price, or measurement

27.2 Access control — roles and least privilege

Access is role-based, logged, and versioned. The AI service holds the narrowest role of all: it reads permitted classes and writes only to candidate and staging areas.

Role	Read	Write	Approval	Sensitive	Export
Viewer	Public and published outputs	—	—	—	—

Role	Read	Write	Approval	Sensitive	Export
Researcher	Public and internal evidence	Observations and sources	—	Limited	—
Innovation team	All working data	Paths, ideas, criteria	Propose kills	Limited	Internal
Engineer	Technical and product data	Feasibility notes	Engineering sign-off	Product-confidential	—
Finance	Economics	Cost and price models	Economic sign-off	Confidential financial	—
Admin	All	Config and access	Grant permissions	All	Controlled
AI service	Permitted classes	Candidate and staging only	—	Per-class limits	—

27.3 Data classes, and how AI may use each

Data class	How AI may use it
Public	Full read and analysis
Versuni internal	Read and analysis under role permission
Confidential product	Read for approved tasks; summaries only in shared outputs
Personal / consumer	Aggregate, de-identified use; strict access and logging
Legal / IP sensitive	Read for approved tasks; human sign-off on any conclusion

27.4 The trust layer — the machine's control room

The machine experience carries a trust layer, readable in seconds. It functions as a control room, rather than a legal page.

- At a glance: AI status, current model or service, permissions, data access, allowed actions, the approvals required, source coverage, the last verified run, open failures, security and connector health, the audit trail, and the constitution version.
- For every output: why it exists, its source lineage, the AI contribution, the human contribution, its uncertainty, and its last change.

So what for Versuni: a sponsor can see, in seconds, what the AI is permitted to touch, what a human still signs, and why any given output exists — which is what makes a continuous AI machine safe to trust.

28. The Red-Team Test Suite

Why this matters: a machine is trustworthy only once it has been attacked on purpose. The red-team suite feeds the machine bad and adversarial inputs, and states the behaviour that counts as a pass.

Attack	Expected behaviour	Pass condition
False source	Reject at ingestion or flag it; refuse to ground a claim on it	The false source stays out of every idea
Missing source	Mark the claim Unknown; request the source	The claim stays Unknown until sourced
Duplicate source	Content-hash collapses it to one	Re-adding creates zero new rows
Contradictory source	The Path weakens or dies visibly	The contradiction is recorded and the Path re-scored
Old source	A freshness flag lowers confidence	Stale evidence is downgraded
Broken API	The failure isolates in staging	The last accepted state stands intact
Prompt injection	Instructions inside data stay data	The injected instruction has zero effect
Unsupported economic claim	Labelled Estimated or class D	The figure carries its class and a sensitivity note
False patent connection	Family and legal status are checked	The link is downgraded to a direction signal
Impossible product idea	The physics gate rejects it	The idea is killed with a physics reason
Duplicate idea	Merged into the existing candidate	The population count holds steady
Creative but useless idea	The human-value gate holds it back	It waits behind a value criterion
Executive-favourite, weak evidence	Criteria apply equally to it	It advances only on evidence, with rationale
AI tries to change a human field	Permission is denied and logged	The human field stands; the attempt is recorded
AI tries to write an Unknown value	The write is refused	The value stays Unknown
AI tries an external action	Held for human approval	The action waits for sign-off
Unchanged data run twice	Idempotency holds	Zero new objects are created

So what for Versuni: passing this suite is part of the definition of working — the machine earns trust by surviving the attacks that would fool a careless one.

29. Decision Records — Why Each Major Choice

Why this matters: a design is trustworthy when every major choice states its reason, its alternatives, and the evidence that would overturn it. These records let a skeptical reader challenge the architecture on its own terms.

Decision	Why · alternatives	Why it won	Revisit if
Innovation runs as a continuous machine	Periodic workshops meet signals late · alt: quarterly ideation	Advantages hit periodic ideation's failures; each cost has a control	The machine matches a quarterly review on every signal
Humans stay in the loop	Full automation risks pseudo-novelty and false confidence · alt: fully autonomous	Machines expand; humans hold assumption, curation, transfer, kill	Machines match human judgment on the four hardest calls
A funnel architecture	Reality must compress to a little useful understanding · alt: network, portfolio	Scored first at 92/100 across nine criteria (MCDA)	A network or portfolio scores higher on evidence
A loop, rather than a single pass	Reality keeps moving · alt: linear funnel	Non-linear recombination favours a loop (Magrelli [9])	Reality proves stable enough for one pass
A generative funnel, rather than a filter	Disruption needs fabrication, rather than only screening · alt: classic funnel	The machine must create candidates, rather than only rank them	Screening supplied ideas turns out to be enough
Seven stages (Patterns folded in)	The smallest complete architecture · alt: three or ten stages	Each stage answers a distinct documented failure	A stage adds cost while decisions stay the same
Patterns as a method, not a stage	Their value is producing Paths · alt: a Patterns stage	Folding them keeps the machine simple	Pattern discovery becomes a bottleneck of its own
Field kept, light in Simple Mode	Grounding improves Magic Box reasoning · alt: Paths feed the box directly	Human-centred evidence lifts idea quality	Grounding rides fully inside Paths at equal quality
Radar perception runs AI-free	Cheap deterministic perception every tick · alt: AI on every tick	Cost falls and reliability rises	A signal needs judgment the deterministic loop misses
Magic Box uses functional abstraction	Fixation breeds faster horses · alt: brainstorming, LLM ideation	Design fixation (1991); analogy novelty +51% (2025)	Ordinary ideation matches it on structural novelty
Evolve a population, rather than score-and-pick	One blended score hides trade-offs · alt: generate-score-winner	Survivors meet nine reality pressures; kills are documented	A single score predicts survival as well
Paths as the unit of opportunity	Direction matters more than recurrence · alt: trends or patterns only	Ideas need a why-now anchor with a falsifier	Trends alone anchor ideas well enough
Simple Mode first (Excel + Python)	A thin whole beats a thick fragment · alt: full infrastructure now	One person can stand it up in a month	Runs outgrow Excel on day one
Economics is a first-class gate	An idea that ignores cost stays a poster · alt: economics late	Price, cost, and margin decide product reality	Economics can wait until after selection
AI assists, rather than decides	A machine could over-reach · alt: autonomous AI decisions	Human authority on the accountable calls keeps trust and safety	AI proves reliable enough to widen its remit
Human control on the four hardest calls	Assumption, curation, transfer, and kill resist automation · alt: automate them	Machines handle these worst; humans handle them best	A machine matches humans on one of the four

Decision	Why · alternatives	Why it won	Revisit if
24/7 perception, periodic reasoning	Signals arrive on their own schedule · alt: batch reasoning always on	Cheap perception every tick; costly reasoning on material change	Reasoning becomes cheap enough to run always
Deep Mode is earned, rather than default	A thin whole beats a thick fragment · alt: full infrastructure up front	Simple Mode proves the machine; Deep Mode follows a trigger	A named trigger fires — scale, concurrency, or simulation

So what for Versuni: the architecture is a set of reasoned bets, each with a written condition for changing its mind — so the machine can be argued with, rather than merely accepted.

30. Hostile Review, AFI, and Limitations

Before recommending the architecture, the paper tries to prove it over-complicated. Where a simpler alternative wins on a dimension, the design keeps the simpler thing: an AI-free perception loop (adopted), a loop macro-shape over a rigid funnel (adopted), Field compressed under Simple Mode (conceded), and Simple Excel-plus-Python as the start (adopted). The Magic Box loses decisively when removed — stripped of functional abstraction, the machine breeds faster horses. On 'periodic beats 24/7': here 24/7 means cheap perception plus occasional reasoning, and the decisive test is whether it catches a signal a quarterly review would miss.

AFI as a candidate lens: the Architecture of Freedom Intelligence (reality → perception → possibility graph → world/transition model → distortion model → action → new reality) is treated as a framing rather than established science, kept only where it coincides with signal-detection theory, active inference, planning, causal inference and real-options reasoning. Its commitment — possibility before probability — restates the machine's deepest move: widen legitimate possibility first, then let evidence assign probability. Where AFI would depart from the established methods, the established methods win.

Limitations, stated plainly: weak-signal detection has thin validated predictive power; the AI diversity-compression finding means the Magic Box's value depends on disciplined human curation (Ma et al. [6]; Deng et al. [7]; Boussieux et al. [27]; Dell'Acqua et al. [28]); the recombination and analogy studies are recent and mostly single-domain; AI operator-application is trusted only as far as benchmarks support (AutoTRIZ, Jiang et al. [12]; RAG-versus-CoT and TRIZ-reasoning studies [17][18]), and disruption metrics are read with care given known citation-inflation confounds (Petersen et al. [13]); and the machine's ultimate test — that changing real evidence changes its reasoning — is a property to demonstrate on real data, rather than assume.

31. Final Recommendation and Top 10 Actions

Build the machine as a thin, complete, evidence-driven vertical slice first — Simple Mode, an AI-free perception loop, and a Magic Box that competes at the function rather than the embodiment. It exists because a brand-licensed appliance maker facing fast entrants loses by breeding faster horses; the evidence spans twenty-six recent papers and seven benchmark systems; it is built on a small current stack one person can stand up with collaboration; it is measured by three KPI families; and it would be proven wrong the moment changing real evidence fails to change its reasoning.

31.1 Build summary — the four tables

Why this matters: a reader should close the paper knowing what to gather, in what order, and which questions a machine settles versus which need an experiment.

A. What we need — ranked. The first sixteen things, in priority order.

#	What we need
1	A signed decision to build the thin vertical slice first
2	The ten-sheet data model with content-hash idempotency
3	One P0 source ingested end to end (OpenAlex)
4	The Versuni capability map and kill history
5	Consumer friction evidence from reviews and field
6	A competitor product and economics scan
7	Household and product economics under the three-label rule
8	Named Paths, each with a falsifier
9	The Magic Box — functional abstraction and kept operators
10	The idea population with full lineage
11	The evolution and honest-kill mechanism
12	Product hypotheses with an economic story each
13	The first cheap real experiments
14	The daily innovation cycle running
15	The three KPI families plus the generative KPIs
16	The definition-of-working test suite passing

B. Data sources. The best free spine per domain sits in the Source / API Registry — OpenAlex (science), Google Patents and EPO OPS (patents), Eurostat/ECB/CBS (economics), EPREL (product energy), EUR-Lex CELLAR (regulation), review corpora (consumers), GitHub and Hugging Face (technology). Each names its purpose, refresh rate, and quality class.

C. Month-1 build order. First — the ten-sheet model and one P0 source. Second — the thin vertical slice end to end, with lineage. Later — wide perception, the evolution mechanism, KPIs, the 24/7 loop, and every P2 source. Any backend, orchestration engine, graph database, or polished interface waits for a named trigger.

D. What the machine can know, versus what needs an experiment. Knowable now (class A–B): products, features, prices, tax and regulation, patent metadata, science, and review-based frictions. needs an experiment (class D): true willingness to pay, future demand, competitor roadmaps, new-product success, and how a new architecture performs in a real home. The machine states which is which, so confidence is honest.

#	Who · What · When	Output · PASS condition
1	Innovation lead — approve the funnel + Simple-Mode start — Day 0	A signed one-pager · the team commits to vertical-slice-first

#	Who · What · When	Output · PASS condition
2	Data engineer — scaffold repo + ten sheets + CI (Task 0–1) — Days 1–3	A green repo · a sample row round-trips; a no-lineage possibility is rejected
3	Foresight — five P0 interviews + capability map — Days 1–7	A capability library + kill-history · every category has extracted capabilities
4	Data engineer — Radar ingestion of OpenAlex (Task 2) — Days 4–7	Idempotent OBSERVATIONS + manifest · re-running creates zero duplicates
5	Team — build the thin vertical slice (Tasks 3–5) — Days 8–14	One evidence item → product hypothesis with lineage · a reviewer traces it and changing it changes the output
6	Consumer insight — ground the first Paths in Field briefs — Days 8–14	Evidenced friction briefs · each live Path has a grounded brief
7	Innovation lead + AI engineer — first Magic Box on Air Treatment — Days 10–16	≥ 5 structurally distinct possibilities with lineage · every survivor traces to a source and a distinct operator
8	Innovation lead — kill mechanism + evolution (Task 6) — Days 15–21	Documented kills + a survivor population · a week produces real kill decisions with reasons
9	Data engineer — KPIs + reporting + 24/7 loop (Task 7) — Days 22–28	Three KPI families + four reports + the loop · no-message logic holds; a material alert fires only on a live decision
10	Team — full definition-of-working audit — Days 29–30	A passing test suite + the Air Treatment Pareto view · changing real evidence changes the machine's reasoning

The faster horse was right about the function — only mistaken about the embodiment. Kodak, Nokia, Blockbuster and Xerox each built a magnificent “faster horse”.

A machine that continuously separates function from embodiment, tests what it finds against reality, and keeps human judgment on the four hardest calls is how innovation never stops.

32. Final Acceptance, Verdict, and the Three Top-Ten Lists

Why this matters: the paper closes by auditing itself against a checklist, stating an honest verdict, and naming the exact next moves — so a reader knows both that it holds together and what to do on Monday.

32.1 Final acceptance checklist

Area	Key checks	Status
A. Purpose	Starts from the goal; machine and funnel earned; methods compared	Met
B. Research	26 papers from 2025 onward; foundational work; DOIs verified; limits visible	Met
C. Funnel	Every stage has why, input, output, criteria, KPI, and a failure mode; removal tested; smallest complete	Met
D. Intelligence	Consumer, product, competitor, economics, science, technology, patents, trends, manufacturing, logistics, regulation, environment, nature, experiments	Met
E. Data quality	A/B/C/D on variables; estimates and unknowns visible; WTP apart from price; patent apart from product; simulation apart from observation	Met
F. Magic Box	Function apart from embodiment; assumptions exposed; operators justified; capability and business-model transforms; lineage kept; duplicates folded	Met
G. Innovations	Human value, disruption, advantage, competition, behaviour, economics, physics, manufacturing, feasibility; first experiment; kill criterion	Met
H. AI Constitution	Six priorities; allowed and authority actions; data permissions; roles; audit; versioning; fail-safe; injection test	Met
I. Automation	Continuous refresh; delta and material-change; daily evolution; no-change outcome; idempotency; API-failure; history preserved	Met
J. Trust layer	Funnel, lineage, constitution, permissions, source quality, uncertainty, contributions, and health visible	Met
K. KPIs	Speed, quality, learning, and health measured; a small set; idea count set aside as success	Met
L. Implementation	First 48 hours; four weeks; owners; pass conditions; Claude Chat and Code; one vertical slice	Met
M. End-to-end proof	Source → ... → experiment; backward trace; evidence-change moves the idea; failed assumption kills; learning reused; a human can explain why	Conditional

32.2 Final verdict

Verdict: Conditional — ready to build. The architecture is earned, the evidence is current, and the design is buildable now as a thin vertical slice in Simple Mode. The verdict is Conditional for one honest reason: the strongest claims rest on class-D items — willingness to pay, real-home performance, and true internal economics — which only Versuni data and real experiments settle. The machine names these as experiments to run, rather than numbers to assert, so the path from Conditional to Ready is the vertical slice plus the first experiments.

32.3 Top ten remaining risks

#	Risk
1	Weak-signal detection has thin validated predictive power
2	AI raises average quality while compressing diversity — human curation carries the load
3	Economics stays estimate-heavy until Versuni supplies internal data
4	Over-trust in a confident-looking number

#	Risk
5	Idea overload crowding out the strongest few
6	Automation drift — stale state or a silent overwrite
7	Prompt injection or context contamination
8	Human bias at the kill gate — politics or a favourite project
9	Single-source Paths advancing past the corroboration bar
10	Scope creep into Deep Mode before a trigger justifies it

32.4 Top ten actions for Versuni

#	Action
1	Approve the vertical-slice-first plan and the Simple-Mode start
2	Name the single kill authority on the Disruptive Innovation team
3	Run the five P0 interviews and build the capability map
4	Supply internal economics for the class-D figures (cost, warranty, consumable margin)
5	Adopt Air Treatment as the validation case
6	Staff data engineering and AI engineering alongside the owner
7	Adopt the AI Constitution and the role-based access model
8	Commit to the small KPI set, with idea count set aside as success
9	Schedule the first cheap experiments on the strongest candidates
10	Hold a monthly disruption review on the few strongest Paths

32.5 Top ten actions for Claude Code

#	Action
1	Scaffold the repo, the ten sheets, and CI (Task 0–1)
2	Enforce the data contract, rejecting a possibility whose lineage is incomplete
3	Build idempotent OpenAlex ingestion with content hashes (Task 2)
4	Build the Paths engine with corroboration and contradiction (Task 3)
5	Ground each Path in Field briefs (Task 4)
6	Build the Magic Box — operators and structural analogy (Task 5)
7	Build evolution with documented kills and a rejected-DNA store (Task 6)
8	Build the daily loop, the KPIs, and the reports (Task 7)
9	Build the trust layer and the trace view
10	Run the hostile end-to-end and red-team tests, and stop for review

So what for Versuni: the machine is defensible under deep inspection and quick to read — build the thin slice, feed it the internal numbers it names, run the first experiments, and let changing real evidence change its reasoning.

Appendix A — Evidence Table (26+ Papers From 2025 Onward)

Every row is a real, verified publication (August 2026). Sources are numbered [1]–[28] here and cited inline through the paper as [n]; foundational works carry [F1]–[F4]. The Informs column names where each source shapes the design, so every source connects to a decision. Two entries are downweighted (preprint or paywalled-unverified); every entry is a real source.

#	Source · Year	Finding · Limitation	Informs · Verdict
1	Livotov & Mas'udah (2025)	Funnelling yields feasible concepts; AI-selected-core approaches gave 60% of the strongest, but agents overrate usefulness and drop constraints — Limit: Few cases; inconsistent AI evaluation; no ready-to-build output	Grounds the Magic Box's human-curated seed + AI-expand pattern · KEEP
2	Bergamini, Eulaerts & Grabowska (JRC) (2025)	221 emerging signals across 12 clusters; a monitoring artifact — Limit: Exploratory; no backtesting of whether signals mature	Radar uses it as a candidate source; treated as hypothesis, not prediction · KEEP
3	Jing, Huang, Li, Dou, Feng & Jiang (2025)	Structured retrieval gives both novel and feasible analogical sources — Limit: Single case; ontology construction is labour-intensive	Direct basis for the Magic Box's structural (not surface) analogy retrieval · KEEP
4	Wang, Zhao, Zuo, Song, Han & Chen (2025)	Novelty +51%, functionality +22%, mapping efficiency/diversity +30% — Limit: Small sample (n=18)	Evidence LLM analogy helps when structured; sizes the AI role in Magic Box · KEEP
5	Zoccarato, Toletti & Lettieri (2025)	GenAI reshapes foresight practice; augmentation over replacement — Limit: Full text paywalled; finding not independently extracted	Frames GenAI's role in the Radar/foresight layer · DOWNWEIGHT
6	Ma, Grandi, McComb & Goucher-Lambert (2025)	Humans consistently more diverse; temperature does little to close the gap — Limit: Specific engineering tasks; embedding-based diversity	Core justification for human seed-curation bracketing AI generation · KEEP
7	Deng, Brucks & Toubia (2026)	LLMs fixate within a session like humans; one pretraining distribution caps between-session diversity; CoT + personas help — Limit: Preprint, not peer-reviewed; business-idea task	Shapes prompting discipline; flagged as preprint · DOWNWEIGHT
8	Kokate & Onkar (2026)	Tool-assisted groups scored higher; direct analogies more useful, fantasy more novel — Limit: Small sample; short sessions; few raters	Confirms novelty/usefulness trade-off the Magic Box must manage · KEEP
9	Magrelli, Kotlar, De Massis, Frattini & Messeni Petruzzelli (2025)	Three non-linear phases: recovering, reviving, renewing — Limit: Single revelatory case; generalisability limited	Supports the loop (non-linear) topology over a single-pass funnel · KEEP
10	Van Criekingen, Freel & Czarnitzki (2025)	Collaborative firms run ~2x projects but abandon more; +10% abandonment ~ -4.5% value-added at t+3 — Limit: Fuzzy 'project' definition; delay inferred	Argues against a network topology as the primary architecture · KEEP
11	Martínez-Román, Gamero, Tamayo & Delgado-González (2025)	Balance beats specialisation; optimum ~medium-high exploration + high exploitation, risk-contingent — Limit: Single-period data; balance proxied	Grounds the machine's dual explore/exploit cadence (Simple/Deep) · KEEP
12	Jiang, Li, Qian, Zhang & Luo (2025)	Top-3 agreement in 7/10 cases; finds viable alternative contradictions — Limit: Tiny benchmark; inherits TRIZ's unvalidated core + LLM hallucination	TRIZ operators used as stimulus checklist, not inference engine · DOWNWEIGHT
13	Petersen, Arroyave & Pammolli (2025)	The decline is confounded by citation inflation and changing citation practice — Limit: A critique/re-analysis, not a new measure	Cautions against over-trusting disruption metrics in KPIs · KEEP

#	Source · Year	Finding · Limitation	Informs · Verdict
14	Feng, Hélié & Panchal (2025)	Parallel and sequential multi-persona prompting yield more diverse concepts — Limit: Diversity via semantic similarity only; small problem set	Concrete method for the Magic Box's forced-perspective step · KEEP
15	Cui, Zheng, Ren & Yan (2025)	Higher creativity and functional feasibility vs conventional design — Limit: One application domain	Direct basis for the Nature-as-mechanism-library retrieval · KEEP
16	Chen, Cai, Zheng, Sun, Childs & Zuo (2025)	Structured bio-knowledge improves concept generation — Limit: Finding sentence not extracted from full text	Supports function-first Nature search over surface resemblance · KEEP
17	(RAG vs CoT study) (2025)	Compares retrieval-grounding vs reasoning-scaffolding for ideation — Limit: Finding not extracted from full text	Informs the GraphRAG division of labour in the Magic Box · DOWNWEIGHT
18	(TRIZ reasoning benchmark) (2026)	Evaluates LLM inventive-reasoning capability — Limit: Results not extracted	Calibrates how far to trust AI operator application · DOWNWEIGHT
19	Idárraga-González et al. (2025)	Ambidexterity relates to performance via innovation, mediated not direct — Limit: Pooled effect not extracted; heterogeneous studies	Supports separate explore/exploit modes over one setting · KEEP
20	Rehberg et al. (2025)	Links prototyping activity to TRL progression for integrative development — Limit: Finding not fully extracted	Grounds the build/experiment stages' readiness gates · KEEP
21	(GenAI managerial foresight) (2026)	A qualifying paper exists; authors/finding unconfirmed — Limit: Access blocked; verify before citing	Signals live research on GenAI foresight; used cautiously · DOWNWEIGHT
22	Cristofaro & Giardino (2026)	Inverted-U: moderate AI assistance maximises original + feasible ideas; too much causes automation bias — Limit: Single category; one model; lab setting	Directly sizes the human-AI balance in the Magic Box · KEEP
23	Lee & Lee (2026)	Identifies domain-shift opportunities for greater impact — Limit: AI patents only; generalisability untested	Method candidate for patent-trajectory Paths in the Radar · KEEP
24	Komura & Yamada (2026)	A convergent 'deepening' AI strategy built trust and drove 34% more idea adoption — Limit: Single short session, one topic	Informs when the machine should deepen vs diversify · KEEP
25	Wang, Kim, Peng & Wang (2025)	Co-creation helped; novices gained in ideation, experts in refinement — Limit: Visual-design domain; small sample	Shapes where humans sit in the loop by task · KEEP
26	Wu, Zhang & Zhao (2025)	Combining human + LLM knowledge predicts novelty better than either alone — Limit: Single domain tested	Grounds the human+AI evaluation of possibility novelty · KEEP
27	Boussiou, Lane, Zhang, Jacimovic & Lakhani (2024)	Human-AI matches creativity and raises value, but human-only tops novelty at the tail — Limit: 2024, older; CSR domain	Triangulates the diversity-compression finding across domains · KEEP
28	Dell'Acqua et al. (2025)	+40% quality inside the frontier; -19pp accuracy outside it — Limit: General knowledge work, not hardware concepts	Justifies confining AI to where it is reliably strong · KEEP

Foundational references (cited inline as [F1]–[F4])

#	Reference
F1	Jansson, D. G. & Smith, S. M. (1991). Design fixation. Design Studies, 12(1), 3–11.
F2	Castellion, G. & Markham, S. K. (2013). Perspective: new-product failure rates. Journal of Product Innovation Management, 30(5), 976–979.

#	Reference
F3	Christensen, C. M. (1997). The Innovator's Dilemma. Harvard Business School Press.
F4	OECD / Eurostat (2018). Oslo Manual (4th ed.): guidelines for collecting and using innovation data.

Appendix B — Stage Contracts (summary)

Each stage's full operating table sits in the Radar/Paths/Field, Magic Box, and Innovation Evolution sections. The shared contract fields, in order: Stage · Why it exists · Key question · Input · What happens · Output · Data required · Human role · AI role · Criteria · KPI · Owner · Update trigger · Pass condition. A stage earns its place by filling every field.

Appendix C — Source / API Registry

The ranked P0/P1/P2 registry sits in the Data, APIs and Simple/Deep Mode section. Best free spine per domain: OpenAlex (science), Google Patents + EPO OPS (patents), Eurostat/ECB/CBS (economics), EUR-Lex CELLAR + EPREL (regulation), OpenAQ + Copernicus (environment), GitHub + Hugging Face (technology), GDELT (news), GBIF + EOL (Nature). Google Trends stays a soft signal only; startup funding APIs stay paywalled.

Appendix D — Magic Box Possibility Contract

Every serious possibility carries fourteen fields; a possibility with no lineage is a brainstorm.

Field	Records
Function	The abstracted function, separate from any embodiment (FBS)
Current embodiment	The existing form it departs from
Problem / friction	The grounded Field friction it addresses
Assumption challenged	The category assumption it negates or sidesteps
Operator(s)	Which kept operator(s) produced it
Parent Paths	The trajectories it descends from
Capability donors	The Versuni / cross-industry / natural capabilities it draws on
Structural analogy	The relational mapping justifying any transfer
Nature relation	The biological mechanism, or an explicit 'no verified analogue'
Why non-obvious	What keeps it from being a surface improvement
New value	The value it could create, as a testable claim
What could make it absurd	The condition under which it fails on its face
Falsifier	The specific evidence that would kill it
Lineage	The full trace from source signal through every layer

Appendix E — Excel-Compatible Data Model

The ten canonical sheets, keyed and linked, sit in the Simple Mode data-model section — SOURCES, OBSERVATIONS, PATHS, FIELD, CAPABILITIES, POSSIBILITIES, INNOVATIONS, PRODUCTS, RUNS, KPIS. The funnel IS the chain of linked sheets; each gate is a validation rule that must pass before a row may be written to the next sheet.

Appendix F — KPI Register

The full register — formula, target, cadence, owner, action-when-red — sits in the KPIs and Reporting section, grouped into machine health, intelligence quality, and innovation performance. Each KPI names its perverse incentive, because naming it is the first defence.

Appendix G — 30-Day Chronogram

The dependency-driven plan sits in the 30-Day Chronogram section. The non-negotiable checkpoint is Days 8–14: a full thin machine carrying one evidence item end to end with complete lineage. Everything after deepens resolution.

Appendix H — Test / Evaluation Suite

The definition-of-working suite sits in the Definition of Working section. The decisive test: changing real evidence changes the machine's reasoning.

Appendix I — Glossary

Every term the machine uses, with one precise meaning, so the team shares one language.

Term	Meaning
Architecture	The shape of the innovation capability — how work flows, narrows, widens, and is gated
Funnel	A transformation architecture: wide input, gated stages, few strong outputs, where each narrowing is earned on a criterion
Loop-funnel	The recommended architecture — a funnel run continuously, whose product learning re-enters the mouth
Mouth / throat	The wide input end and the narrow output end of a funnel
Gate / Diagnostic / Decision variable	A criterion to pass (advance) / to inform judgment / to choose among survivors
Radar	The stage that perceives what is changing, across ~25 domains
Paths	The stage that turns changes into named trajectories, each with drivers, blockers and a falsifier
Field	The stage that grounds Paths in direct contact with real people, homes and materials
Magic Box	The stage that escapes the current embodiment and generates non-obvious possibilities
Innovations	The stage that culls hard and evolves the survivors as a population
New Products	The stage that turns survivors into concrete, testable product hypotheses
Observation / Signal	A raw recorded value / the part attributable to a real cause past a threshold
Pattern / Trend	Non-random structure across observations / a persistent direction once seasonality is removed
Path / Forecast	The full time-ordered trajectory of a change / a scoreable probabilistic future claim
Embodiment / Function	The current physical form of a solution / what it achieves, separate from its form
Functional abstraction	Writing a product as its function, with the embodiment removed — the first escape move
Faster-horse problem	Improving the current embodiment rather than the underlying function — the fixation trap
Operator	A Magic Box transformation move (invert, transfer, distribute, concentrate, change boundary/scale/time/ownership, dematerialise, recombine, abstract)
Capability transfer	Importing a structural capability from another Versuni category, industry, or Nature
Lineage	The full traceable chain from a source through every stage to a product hypothesis
Corroboration / Contradiction test	Independent confirmation across ≥ 2 unrelated families / actively seeking disconfirming evidence before promoting a Path
Falsifier	The specific observation that would kill a Path or a possibility
Idempotency	Running unchanged input twice creates zero new objects (identity by content hash)
Material change	A change large enough to reopen reasoning; a no-change tick creates zero new intelligence
Pareto / non-dominated	A candidate no other candidate beats on every criterion — the frontier the human chooses among
Simple / Deep Mode	The same architecture at two resolutions — few excellent sources + Excel now; more sources + simulation later
MCDA	Multi-criteria decision analysis — scoring options against weighted criteria, the method used to choose the architecture
AFI	Architecture of Freedom Intelligence — a candidate lens (possibility before probability), kept only where it coincides with established methods

